

Book II of Euclid's Elements

Formal Reconstruction — Technical Documentation

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Document Status

This volume is the technical documentation on formal reconstruction of Book II of Euclid's *Elements*. It records the mathematical architecture of the reconstruction, the Lean formalization, the dependency structure of individual propositions, the reusable rectangle and scissors-congruence infrastructure, and the relation between the formal development and the classical sources.

The documentation accompanies the formal development. The Lean source is authoritative for theorem statements, hypotheses, dependencies, and proof status.

Contents

Document Status	ii
List of Figures	ix
1 Proposition II.1	1
1.1 Introduction	1
1.2 Euclid’s “Rectangle Contained By”	2
1.2.1 The original geometric meaning	2
1.2.2 Why this already looks algebraic	2
1.3 The Synthetic Rectangle Layer	2
1.3.1 Rectangles	2
1.3.2 The formal meaning of “contained by”	3
1.3.3 No segment multiplication and no numerical area	3
1.4 Exact Scissors Equality and Equal Content	4
1.4.1 Exact scissors equality	4
1.4.2 The more general equal-content relation	4
1.5 The Classical and Formal Configurations	4
1.5.1 The binary split	4
1.5.2 The three-part split	5
1.5.3 Scope of the present formal statement	6
1.6 From a Concrete Diagram to Euclid’s Language	6
1.7 Representative Independence	6
1.7.1 Why representatives matter formally	6
1.8 Commutativity by Rectangle Rotation	7
1.9 Proof Architecture of the Reconstruction	8
1.9.1 Step 1: exact binary dissection	8
1.9.2 Step 2: iterate the split	8
1.9.3 Step 3: attach the “contained by” data	9
1.9.4 Step 4: remove the choice of representatives	9
1.9.5 Step 5: expose the symmetric side order	9
1.10 Classical Proof and Formal Proof	9
1.10.1 Euclid’s proof pattern	9
1.10.2 The formal DAG	10
1.11 Geometry, Content, and Representation	10
1.11.1 Geometry	10
1.11.2 Content	10
1.11.3 Representation	10
1.12 Hidden Configuration Data	11
1.13 Axiomatic Status	11
1.14 Relationship with Book I and the Following Propositions	12
1.15 Source Audit	12
1.15.1 Euclid	12

1.15.2	Hartshorne	12
1.15.3	Hilbert	12
1.15.4	The phrase “geometric algebra”	13
1.16	Dependency Summary	13
1.17	Conclusion	14
2	Proposition II.2	16
2.1	Introduction	16
2.2	Euclid’s Statement and Classical Configuration	16
2.2.1	The statement	16
2.2.2	The classical diagram	17
2.3	Classical Proof	17
2.4	The Geometric Identity	18
2.5	Reusable Book II Infrastructure	18
2.5.1	A square as a rectangle contained by the same side twice	18
2.5.2	The binary form of II.1	19
2.5.3	Exchanging the containing-side order	19
2.6	Proof Architecture of the Reconstruction	20
2.6.1	Step 1: recognize the square	20
2.6.2	Step 2: rotate the two arbitrary part representatives	20
2.6.3	Step 3: apply II.1 with fixed segment BC	20
2.6.4	Step 4: return to Euclid’s ordering	21
2.7	Lean Formalization	21
2.8	Relationship with Proposition II.1	22
2.9	Relationship with Book I Infrastructure	23
2.10	Formalization Notes and Hidden Geometric Data	23
2.11	Geometry, Content, Representation, and Later Arithmetic	24
2.11.1	Geometry	24
2.11.2	Content / scissors	24
2.11.3	Representation	24
2.11.4	Later arithmetic / numerical area	24
2.12	Axiomatic and Semantic Status	24
2.13	Hartshorne’s Semantic Control	25
2.14	Hilbert’s Layering	25
2.15	Dependency Summary	25
2.16	Conclusion	26
3	Proposition II.3	27
3.1	Introduction	27
3.2	Euclid’s Statement and Classical Configuration	27
3.2.1	The statement	27
3.2.2	The classical diagram	28
3.3	Classical Proof	28
3.4	The Geometric Identity	30
3.5	Reusable Book II Infrastructure	30
3.5.1	The binary form of II.1	30
3.5.2	A square as $\text{Rect}(MC, MC)$	31
3.5.3	What II.3 does not use	31
3.6	Proof Architecture of the Reconstruction	31
3.6.1	Step 1: recognize the square	31
3.6.2	Step 2: instantiate II.1 with the fixed segment MC	32
3.7	Lean Formalization	32

3.8	Relationship with Earlier Book II Propositions	34
3.8.1	Proposition II.1	34
3.8.2	Proposition II.2	34
3.9	Relationship with Book I Infrastructure	34
3.10	Formalization Notes and Hidden Geometric Data	35
3.11	Geometry, Content, Representation, and Later Arithmetic	35
3.11.1	Geometry	35
3.11.2	Content / scissors	35
3.11.3	Representation	35
3.11.4	Later arithmetic / numerical area	36
3.12	Axiomatic and Semantic Status	36
3.13	Hartshorne's Semantic Control	36
3.14	Hilbert's Layering	37
3.15	Dependency Summary	37
3.16	Conclusion	38
4	Proposition II.4	39
4.1	Introduction	39
4.2	Euclid's Statement and Classical Configuration	40
4.2.1	The statement	40
4.2.2	The classical diagram	40
4.3	Classical Proof	40
4.4	The Geometric Identity in the Current Formalization	42
4.5	Reusable Book II Infrastructure	42
4.5.1	Proposition II.2: split the square on the whole	42
4.5.2	Proposition II.3: expand a whole-by-part rectangle	43
4.5.3	Rectangle rotation and endpoint reversal	43
4.5.4	Exact scissors addition	44
4.6	Proof Architecture of the Reconstruction	44
4.6.1	Step 1: reverse the cut orientation for the left branch	44
4.6.2	Step 2: apply II.2 to the square on the whole	44
4.6.3	Step 3: prepare the reversed left II.3 call	44
4.6.4	Step 4: expand the left rectangle by II.3	45
4.6.5	Step 5: expand the right rectangle by II.3	46
4.6.6	Step 6: add the branches and attach them to the square	46
4.7	Lean Formalization	48
4.8	Relationship with Earlier Book II Propositions	51
4.8.1	Proposition II.1	51
4.8.2	Proposition II.2	51
4.8.3	Proposition II.3	51
4.9	Relationship with Book I Infrastructure	51
4.10	Formalization Notes and Hidden Geometric Data	52
4.11	Geometry, Content, Representation, and Later Arithmetic	52
4.11.1	Geometry	52
4.11.2	Content / scissors	52
4.11.3	Representation	53
4.11.4	Later arithmetic / numerical area	53
4.12	Axiomatic and Semantic Status	53
4.13	Hartshorne's Semantic Control	54
4.14	Hilbert's Layering	54
4.15	Dependency Summary	55
4.16	Conclusion	55

5	Proposition II.5	57
5.1	Introduction	57
5.2	Euclid’s Statement and Classical Configuration	57
5.2.1	The statement	57
5.2.2	The classical diagram	58
5.3	Classical Proof	58
5.4	The Geometric Identity in the Current Formalization	59
5.5	Reusable Book II Infrastructure	59
5.5.1	The binary rectangle split from II.1	59
5.5.2	Proposition II.3	60
5.5.3	Rectangle representation and transport	60
5.6	Proof Architecture of the Reconstruction	60
5.6.1	Step 1: recover the full order and midpoint congruence	60
5.6.2	Step 2: apply II.3 on the left half	60
5.6.3	Step 3: transport the midpoint side	61
5.6.4	Step 4: split the unequal part DB	61
5.6.5	Step 5: the scissors bridge	61
5.6.6	Step 6: assemble the right side	61
5.6.7	Step 7: recognize the half-square	61
5.7	Lean Formalization	62
5.7.1	The public theorem signature	62
5.7.2	The final three-stage assembly	64
5.8	Relationship with Earlier Book II Propositions	66
5.8.1	Proposition II.1	66
5.8.2	Proposition II.3	66
5.8.3	Proposition II.4	66
5.9	Relationship with Book I Infrastructure	66
5.10	Formalization Notes and Hidden Geometric Data	67
5.10.1	Order orientation	67
5.10.2	Midpoint as geometry, not arithmetic	67
5.10.3	Representative independence	67
5.10.4	No gnomon primitive	67
5.10.5	No cancellation axiom	67
5.11	Axiomatic and Semantic Status	67
5.12	Dependency Summary	68
5.13	Conclusion	69
A	Greenberg: Foundational and Semantic Control	70
.1	Euclid I.1–I.20 in Greenberg’s Hilbert-Style Development	70
.1.1	Purpose of the comparison	70
.1.2	Comparison table	70
.1.3	Result of the query	72
.1.4	Consequences for the formal reconstruction	72
.1.5	Current checkpoint	72
A	Hartshorne: Equal Content and the Algebra of Areas	73
A.1	Purpose and scope	73
A.2	Hartshorne’s organization of the problem	73
A.3	Comparison table	74
A.4	Foundational observations retained from Hartshorne	77
A.5	Result of the Hartshorne audit	77
A.6	Current checkpoint	78

A.7	Book II semantic checkpoint: Proposition II.5	78
B	Hilbert: Content and Segment Arithmetic	79
B.1	Purpose and scope	79
B.2	Hilbert's organization of the relevant theory	79
B.3	Comparison table	79
B.4	Structural observations retained from Hilbert	80
B.5	II.1–II.5: exact dissection below segment arithmetic	81
B.6	Result of the Grundlagen audit	81
B.7	Current checkpoint	82
C	Borsuk–Szmielw: Axiomatic Layers and Representation	83
C.1	Why this source matters	83
C.2	Layer notation	84
C.3	Euclid in the Borsuk–Szmielw development	84
C.4	The strongest architectural correspondences	89
C.4.1	Transport is primitive-strength infrastructure	89
C.4.2	Pairs of Euclidean propositions become biconditionals	89
C.4.3	Midpoints and bisectors include positional information	89
C.5	The neutral parallel theory is already large	90
C.6	Theorem 12 of Chapter V as an interface theorem	90
C.7	What the book contributes at I.35	91
C.8	Consequences for the formal library	91
C.8.1	Hilbert interface candidates	92
C.8.2	Book Zero candidates	92
C.8.3	Euclidean interface candidates	93
C.9	Conclusion	93
	Bibliography	94

List of Figures

1.1	Binary and three-part rectangle dissections	5
1.2	Commutativity from cyclic rectangle rotation	8
2.1	Square dissection and rectangle rotation	17
3.1	Binary rectangle split with the square on MC	29
4.1	Square decomposition into two squares and two equal rectangles	41
5.1	The gnomon mechanism in II.5	58

Chapter 1

Proposition II.1

1.1 Introduction

Proposition II.1 opens Book II with a change of mathematical language. Book I has developed a synthetic theory of congruence, parallels, parallelograms, squares, and equal content. Book II begins to combine these ingredients in a systematic calculus of rectangles. A modern reader immediately recognizes in II.1 the pattern of a distributive law. That recognition is useful, but it must be stated carefully: Euclid does not introduce variables for numerical lengths, does not define multiplication of lengths, and does not prove an identity in a number field.

The primitive object in the proposition is a geometric figure: a rectangle “contained by” two straight lines. When one of the two lines is divided into parts, the large rectangle is divided into smaller rectangles having the same other side. The equality asserted by the proposition is therefore first of all a statement about decomposition of figures.

This is a natural point at which to speak of the beginning of the *geometrical algebra* characteristic of Book II. The phrase is an interpretive description of a stable pattern of geometric operations, not a claim that Euclid was secretly manipulating modern symbolic formulas. In the present formal reconstruction this distinction is especially sharp: the code contains rectangles, segment congruence, betweenness, parallelograms, right angles, and scissors relations, but no multiplication operation on segment lengths and no numerical area function.

The formal development separates three levels:

1. a concrete rectangle and its geometric cuts;
2. the Euclidean phrase “rectangle contained by” two given segments;
3. independence of the particular rectangle chosen to represent that phrase.

The final theorem is stronger than a bare equal-content statement. Its conclusion is the exact relation

```
HilbertScissorsEq
```

for the whole rectangle and the formal sum of its parts.

1.2 Euclid’s “Rectangle Contained By”

1.2.1 The original geometric meaning

Book II begins with the terminology that a rectangular parallelogram is “contained by” the two straight lines which contain one of its right angles. Thus the expression does not initially denote a product of two numbers. It names a rectangle by two adjacent sides.

Suppose a rectangle has adjacent sides congruent to two given segments PQ and RS . Euclid’s phrase

the rectangle contained by PQ, RS

should be read geometrically: choose a right-angled parallelogram whose two adjacent sides have those magnitudes. The rectangle itself remains a plane figure.

For exposition only, we will occasionally write

$$\text{Rect}(PQ, RS).$$

This is documentary shorthand. It is not a Lean definition, it is not a number-valued function, and the comma is not multiplication.

1.2.2 Why this already looks algebraic

If BC is divided into consecutive parts BM , MN , and NC , II.1 says in this shorthand

$$\text{Rect}(BC, PQ) = \text{Rect}(BM, PQ) + \text{Rect}(MN, PQ) + \text{Rect}(NC, PQ).$$

A modern algebraist naturally sees the shape

$$(b_1 + b_2 + b_3)a = b_1a + b_2a + b_3a.$$

The formal proof, however, establishes the geometric statement on the first line, not the numerical identity on the second. The second formula is only a modern mnemonic for the combinatorial pattern of the dissection.

This is consistent with the modern synthetic reading of Euclid’s area theory. Hartshorne treats Book II as a collection of equal-content statements about figures and describes the resulting manipulation as “geometrical algebra.” The point is precisely that the manipulation precedes any assignment of numerical areas to the figures.

1.3 The Synthetic Rectangle Layer

1.3.1 Rectangles

The reconstruction introduces a small reusable rectangle API in `HilbertRectangle.lean`. A rectangle is a parallelogram with one specified right angle:

```
def IsRectangle
  (A B C D : Geo.Point) : Prop :=
  IsParallelogram Geo A B C D /\
  HilbertRightAngle Geo A B C
```

The parallelogram structure supplies the opposite-side relations, while the right-angle component distinguishes rectangles from arbitrary parallelograms. The remaining right angles can then be propagated synthetically.

1.3.2 The formal meaning of “contained by”

The Euclidean phrase is represented by the predicate

```
def IsRectangleContainedBy
  (A B C D P Q R S : Geo.Point) : Prop :=
  IsRectangle Geo A B C D /\
  Geo.Congruent A B P Q /\
  Geo.Congruent B C R S
```

Thus

```
IsRectangleContainedBy Geo A B C D P Q R S
```

means exactly that $ABCD$ is a concrete rectangle, $AB \cong PQ$, and $BC \cong RS$. Nothing in this definition assigns a number to PQ or RS . The two given segments specify the two adjacent geometric magnitudes of a representative rectangle.

The layer also proves existence of such a representative and, more importantly for the semantics of Book II, uniqueness of its scissors content. The theorem

```
rectangle_contained_by_unique
```

says that any two rectangles contained by the same ordered pair of segments satisfy the exact scissors relation. Consequently Euclid’s definite description “the rectangle contained by PQ, RS ” becomes well-defined at the scissors level even though many different concrete quadrilaterals can represent it.

1.3.3 No segment multiplication and no numerical area

There are two tempting modernizations that the present reconstruction explicitly avoids.

First, we do not define a product of segment lengths $|PQ| |RS|$. The pair (PQ, RS) remains a pair of geometric segments, and the rectangle is related to that pair by congruence of adjacent sides.

Second, we do not map a rectangle to a real number called its area. The content of the proposition is expressed through relations between finite formal sums of triangles. This permits addition, dissection, transport, and comparison of figures at the synthetic level.

This is also conceptually distinct from Hilbert’s later arithmetic of segments, where arithmetic operations are introduced as an additional construction. No such field or segment-arithmetic layer is imported into II.1.

1.4 Exact Scissors Equality and Equal Content

The project uses two related but different content relations. They should not be conflated.

1.4.1 Exact scissors equality

`HilbertScissorsEq` records an exact finite cut-and-paste relation between formal scissors terms. In II.1 it is produced by explicit splits of the rectangle into smaller rectangles and by normalization of the triangulated parallelogram terms. Schematically,

$$P \simeq_{\text{sc}} Q + R$$

means that the two sides can be matched by the exact scissors machinery already developed in the project.

The rectangle transport theorem is already available in this stronger form:

```
rectangle_transport_scissorsEq
```

for rectangles with congruent corresponding adjacent sides.

1.4.2 The more general equal-content relation

The broader project notion corresponding to equal content is

```
HilbertScissorsEquicomplementable
```

It allows suitable complementary terms to be adjoined before exact scissors equality is established. Thus it is designed to capture a more general synthetic equal-content relation.

The theorem

```
rectangle_transport
```

is obtained by weakening the exact transport theorem from `HilbertScissorsEq` to equicomplementability.

For II.1 no such weakening is necessary. The final theorem proves `HilbertScissorsEq` itself. Therefore II.1 certainly yields an equal-content consequence, but its formal statement retains the stronger fact that the displayed equality comes from an exact dissection.

1.5 The Classical and Formal Configurations

1.5.1 The binary split

The reusable core is a one-cut theorem. Let $B - M - C$ and let the corresponding point L divide the opposite side DE . The rectangle $BCED$ is cut into the rectangles $BMLD$ and $MCEL$:

$$BCED \simeq_{\text{sc}} BMLD + MCEL.$$

In Lean the theorem is

```
rectangle_split
```

and it returns both the rectangle structure of the two pieces and the exact scissors equality.

The formal configuration contains one additional point X . It satisfies $D - X - C$ and $M - X - L$. This is not decorative data: the inherited exact split lemma from the I.47 scissors infrastructure uses the intersection of the cut line with a diagonal. The classical rectangle picture suppresses this witness; the formal theorem makes it explicit.

1.5.2 The three-part split

The three-part form applies the binary split twice. First cut at $M - L$; then cut the remaining rectangle at $N - K$. The result is

$$\begin{aligned} BCED &\simeq_{\text{sc}} BMLD \\ &\quad + MNKL \\ &\quad + NCEK. \end{aligned}$$

The second split has its own diagonal-intersection witness Y , with $L - Y - C$ and $N - Y - K$.

Figure 1.1 shows both levels and includes the hidden witnesses X and Y required by the current Lean interface.

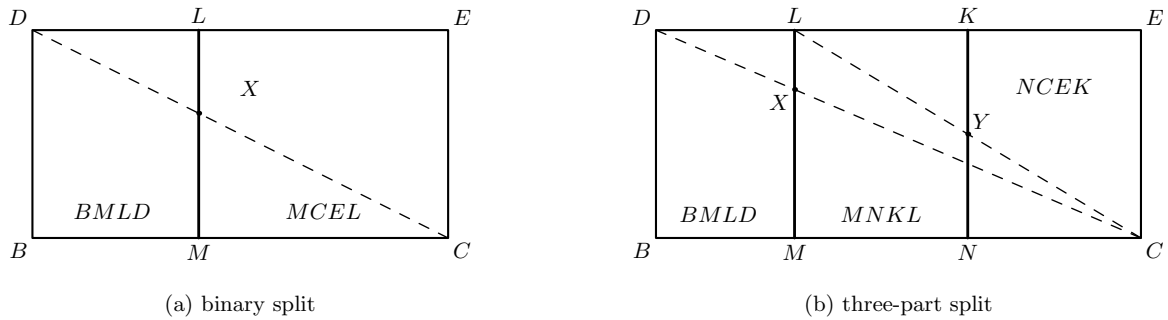


Figure 1.1: The exact rectangle splits underlying II.1. Panel (a) is the binary kernel: $B - M - C$ and the cut ML divide $BCED$ into $BMLD$ and $MCEL$; the dashed diagonal DC meets the cut at the formal witness X . Panel (b) applies the same construction twice: BC is divided into BM , MN , and NC , producing the three rectangles $BMLD$, $MNKL$, and $NCEK$. The dashed diagonals display the witnesses X and Y used by the two exact split calls.

The actual three-part theorem is

```
rectangle_split_three
```

Its proof is structurally transparent: obtain the first binary equality, split the remaining right-hand rectangle, add the unchanged left piece to that second equality, and compose by transitivity of `HilbertScissorsEq`.

1.5.3 Scope of the present formal statement

Euclid states II.1 for one line cut into an arbitrary finite number of parts. The current Lean file does not yet package an n -ary list-valued theorem. It proves the binary split and the explicit three-part form used in the standard II.1 configuration. The binary theorem is the reusable kernel from which a general finite version could later be obtained by iteration.

This is a deliberate statement-fidelity point. The documentation records Euclid’s general formulation, but the final formal statement displayed below is the representative-independent three-part theorem actually present in `Proposition2_1.lean`.

1.6 From a Concrete Diagram to Euclid’s Language

The geometric split alone produces concrete rectangles. Euclid II.1 is more abstract: it speaks of the rectangles *contained by* the whole segment and the several parts together with one fixed segment.

The theorem

```
rectangle_split_three_contained_by
```

bridges these levels. If $CE \cong PQ$, it packages the four concrete rectangles as

$$\begin{aligned} BCED & : \text{Rect}(BC, PQ), \\ BMLD & : \text{Rect}(BM, PQ), \\ MNKL & : \text{Rect}(MN, PQ), \\ NCEK & : \text{Rect}(NC, PQ). \end{aligned}$$

The opposite-side congruences of the parallelograms are what transport the fixed side PQ through the two cuts. The conclusion still contains the same exact scissors equality between the concrete quadrilateral terms.

At this stage the geometry corresponding to Euclid’s wording is visible, but the theorem still uses one particular drawn rectangle and its particular cut points. The final step removes this representational dependence.

1.7 Representative Independence

1.7.1 Why representatives matter formally

A phrase such as

$$\text{Rect}(BC, PQ)$$

does not identify four vertices. Lean, however, must manipulate an actual term such as

```
hilbertParallelogramTerm Geo U0 U1 U2 U3
```

with named vertices. The final theorem therefore accepts arbitrary representatives for the whole rectangle and for each of the three part rectangles.

The concrete dissection first proves

$$BCED \simeq_{\text{sc}} BMLD + MNKL + NCEK.$$

Then four calls to

```
rectangle_contained_by_unique
```

transport the whole and the three pieces to arbitrary representatives U , V , W , and Z . Compatibility of `HilbertScissorsEq` with formal addition transports the nested sum. Transitivity completes the chain

arbitrary whole representative
 $\Downarrow$
 concrete whole rectangle
 $\Downarrow$
 concrete three-part dissection
 $\Downarrow$
 arbitrary representatives of the three parts.

This is the formal content of representative independence. It is not merely a convenience for Lean. It explains why the Euclidean expression “the rectangle contained by two segments” can function as a stable geometric magnitude without first quotienting segments into numerical lengths or assigning numerical areas to rectangles.

1.8 Commutativity by Rectangle Rotation

The second algebra-like feature is symmetry in the two containing segments. For a numerical product one would write $ab = ba$. The synthetic proof does not appeal to a commutative multiplication law. Instead it rotates the presentation of a rectangle.

If $ABCD$ is contained by PQ, RS , then the cyclically reoriented rectangle $BCDA$ has adjacent sides BC and CD . The first is already congruent to RS ; the second is congruent to AB , hence to PQ . Thus the same geometric rectangle, with its vertices read from the next corner, is contained by RS, PQ .

The exact theorem available in the rectangle layer is

```
rectangle_contained_by_swap
```

It returns both the swapped `IsRectangleContainedBy` fact and a `HilbertScissorsEq` identifying the two quadrilateral presentations. The current rectangle API also packages the representative-independent consequence as

```
rectangle_contained_by_comm
```

so the commutativity needed by later Book II propositions is a theorem of rectangle geometry rather than an assumed law of segment multiplication.

Figure 1.2 shows the geometric mechanism. The operation is a cyclic change of the distinguished corner of the same rectangle; this exchanges the two adjacent side roles.

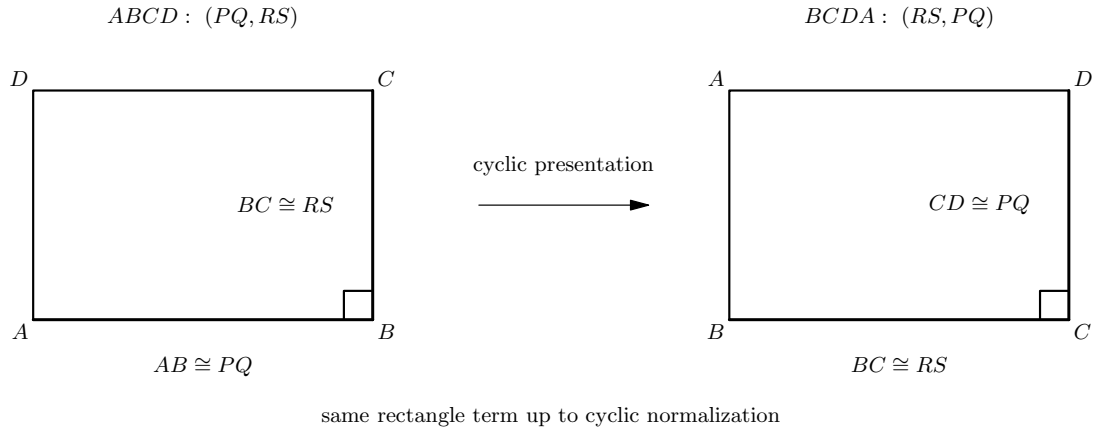


Figure 1.2: The geometric source of commutativity. On the left, $ABCD$ represents the rectangle contained by PQ, RS . Reading the same rectangle from the next vertex gives $BCDA$: now the first adjacent side represents RS and the second represents PQ . Exact scissors invariance under this cyclic presentation change yields the synthetic counterpart of exchanging the two factors.

This lemma is already operational in the next proposition. II.2 naturally writes rectangles in the order “whole side, part”, whereas the II.1 cut lemma produces them in the order “part, fixed side”. The implementation of II.2 uses the swap theorem to pass between these two presentations before applying the binary form of II.1.

1.9 Proof Architecture of the Reconstruction

The Lean proof is best read as five mathematical stages.

1.9.1 Step 1: exact binary dissection

```
rectangle_split_scissorsEq
rectangle_split_left
rectangle_split_right
rectangle_split
```

The inherited I.47 split theorem supplies the exact scissors decomposition. The two local rectangle helpers prove that each piece remains a rectangle by propagating right angles through the relevant parallelograms and along the same rays determined by betweenness.

1.9.2 Step 2: iterate the split

```
rectangle_split_three
```

The right-hand remainder of the first split is itself a rectangle, so the same binary theorem can be applied again. Addition and transitivity of exact scissors equality compose the two decompositions.

1.9.3 Step 3: attach the “contained by” data

```
rectangle_split_three_contained_by
```

Opposite-side congruence transports the fixed segment PQ through all three pieces. Each concrete piece is thereby promoted from `IsRectangle` to `IsRectangleContainedBy`.

1.9.4 Step 4: remove the choice of representatives

```
rectangle_contained_by_unique
euclid_proposition_2_1_three
```

The uniqueness theorem replaces the concrete whole and each concrete part by an arbitrary rectangle contained by the same pair of segments. This is the semantic step that makes the final statement faithful to Euclid’s definite phrase “the rectangle contained by.”

1.9.5 Step 5: expose the symmetric side order

```
rectangle_contained_by_swap
rectangle_contained_by_comm
```

Cyclic rotation of the rectangle exchanges the two containing segments. This is not needed to establish the displayed three-part split itself, but it is an important part of the Book II rectangle interface and is immediately useful in II.2 and later propositions.

1.10 Classical Proof and Formal Proof

1.10.1 Euclid’s proof pattern

In the classical three-part configuration, take one undivided segment A and a segment BC divided at D and E . Erect a perpendicular at one endpoint of BC , lay off on it a segment equal to A , complete the outer rectangle, and through the division points draw parallels to the perpendicular side. The outer rectangle is visibly partitioned into three rectangles. Each smaller rectangle is contained by A and one of the three consecutive parts of BC . Common Notion 2 then combines the parts into the whole equality.

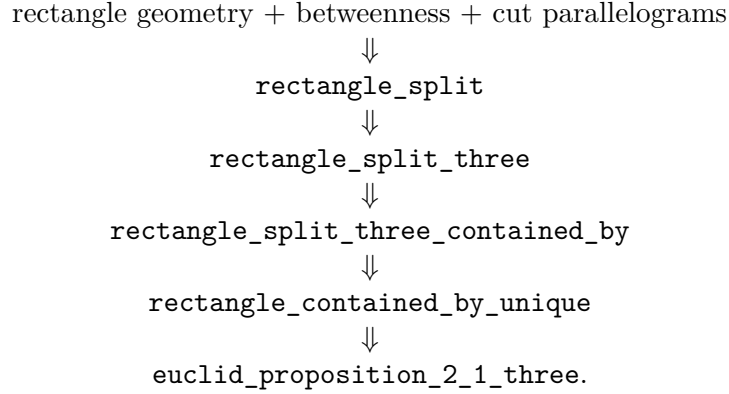
The classical DAG is therefore extremely small:

```

      divide one side
        ↓↓
    draw the corresponding parallels
        ↓↓
    obtain three subrectangles
        ↓↓
whole rectangle = sum of the three subrectangles.
```

1.10.2 The formal DAG

The current reconstruction expands two things that the classical diagram treats as immediate: exact scissors bookkeeping and independence of the chosen rectangle representatives.



The proof is therefore not a mechanical transcription of Euclid. It preserves the geometric core, but makes the semantic and representational layers explicit.

1.11 Geometry, Content, and Representation

1.11.1 Geometry

The geometric data are substantial even though the proposition is visually simple. The current theorem receives the outer rectangle, strict betweenness relations on the divided side and its opposite, two diagonal-intersection witnesses, and four parallelogram facts for the two successive cuts. Right angles of the two resulting rectangles are not read from the picture; they are proved by right-angle transport and parallelogram geometry.

No coordinates are introduced. The cut points are controlled by **Between**, the side lengths by segment congruence, and the rectangular structure by parallels and synthetic right angles.

1.11.2 Content

Content enters in its strongest local form. The core statements are exact **HilbertScissorsEq** decompositions. The three-part theorem is composed from two exact binary splits by addition and transitivity. Representative transport is also exact scissors equality.

The more general equicomplementability relation is part of the surrounding Book I content theory, but II.1 does not need to weaken its conclusion to that level.

1.11.3 Representation

Representation is central to the formal meaning of the proposition. A rectangle contained by PQ, RS is not one canonical four-tuple of vertices. The predicate **IsRectangleContainedBy** describes a representative; **rectangle_contained_by_unique** proves that different representatives have the same exact scissors content.

There is a second representation issue inside the split itself. The inherited scissors theorem and the Book II statement use different cyclic orders for some parallelogram terms. The proof normalizes these by

```
parallelogram_term_rotateOne
```

before composing the scissors equalities. These cyclic changes are bookkeeping of a geometric representation, not new area facts.

1.12 Hidden Configuration Data

The classical drawing makes several facts appear automatic. Lean records them explicitly:

- $B - M - C$ for the first cut and $M - N - C$ for the second;
- $D - L - E$ and $L - K - E$ on the opposite side;
- $D - X - C$ together with $M - X - L$ for the first diagonal/cut intersection;
- $L - Y - C$ together with $N - Y - K$ for the second diagonal/cut intersection;
- explicit parallelogram witnesses for the left and right pieces of each split;
- congruence $CE \cong PQ$, which identifies the fixed side used in the “contained by” descriptions;
- four arbitrary rectangle representatives in the final theorem.

The points X and Y do not appear in Euclid’s statement and need not be emphasized in an elementary picture of II.1. They are nevertheless real data of the current exact-dissection API, inherited from the triangulated proof of the rectangle split.

1.13 Axiomatic Status

The public II.1 theorems assume

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

so II.1 is Euclidean in the current project API. Group IV is available through this typeclass and is consumed transitively by the parallelogram and parallel infrastructure on which the rectangle layer and inherited split theorem depend. The proposition itself does not introduce a new parallel axiom.

No continuity typeclass, Archimedean axiom, numerical area function, or segment field is introduced in `Proposition2_1.lean`. No proposition-local `axiom` or `sorry` occurs in the audited source. The modular builds for `HilbertRectangle` and `Proposition2_1` are recorded as clean at the present project checkpoint.

The content status is also worth separating from the later converse arguments of Book I. II.1 proceeds from geometry to an exact dissection. It does not infer metric geometry from an equal-content hypothesis, so it does not need a proper-part or De Zolt-type strictness argument.

1.14 Relationship with Book I and the Following Propositions

The exact split theorem reuses the scissors infrastructure established during I.47. In particular, the underlying theorem historically named

```
i47_square_split
```

actually requires only parallelogram structure, and II.1 repackages it as a general rectangle split.

The rectangle layer also uses Book I results about parallelograms, opposite sides, congruent right angles, SAS/SSS transport, and the scissors representation of parallelograms. Thus Book II does not start a new foundational geometry; it starts a new way of organizing geometric magnitudes on top of the completed Book I infrastructure.

II.2 immediately imports II.1. Its formal proof illustrates why the symmetric rectangle API matters: Euclid’s order of the two containing sides is opposite to the order produced by the binary cut theorem, so the representatives are swapped by rectangle rotation before II.1 is applied. II.3 continues the same rectangle calculus. In this sense II.1 is not only the first theorem of Book II but the API-level distributive kernel for the propositions that follow.

1.15 Source Audit

1.15.1 Euclid

Book II Definition 1 supplies the semantic key: the two lines “containing” a rectangle are the adjacent sides that contain a right angle. Proposition II.1 then divides one of two given lines into several parts and identifies the whole rectangle with the sum of the rectangles formed from the undivided line and those parts.

The proof is geometric throughout. The division points on one side are carried across the outer rectangle by parallels. The resulting subrectangles literally partition the original rectangle. Nothing in the proposition requires a numerical assignment to a segment or a plane figure.

1.15.2 Hartshorne

Hartshorne’s modern synthetic discussion is a useful semantic control. He interprets Euclid’s equalities of figures through equal content rather than numerical area, and notes that all the Book II propositions are valid at that level. He also uses “geometrical algebra” for the systematic manipulation of such geometric contents.

Our reconstruction is deliberately slightly sharper for II.1: the final result is not only equicomplementability but exact `HilbertScissorsEq`. This is possible because II.1 is itself a direct dissection theorem.

1.15.3 Hilbert

Hilbert is relevant here mainly as a warning against collapsing distinct layers. His later arithmetic of segments introduces arithmetic operations only after substantial synthetic development. The current Book II reconstruction does not import that arithmetic and does not use Hilbert’s

later numerical surface-area theory. It remains in the incidence/order/congruence plus scissors language already available after Book I.

1.15.4 The phrase “geometric algebra”

The phrase is therefore appropriate only with a qualification. II.1 exhibits an algebraic *pattern*: partition of one side corresponds to additive decomposition of a rectangle; exchanging the two containing sides corresponds to cyclic rotation; a square is the special case in which the two containing segments are the same. These operations anticipate the formal laws of a product-like calculus.

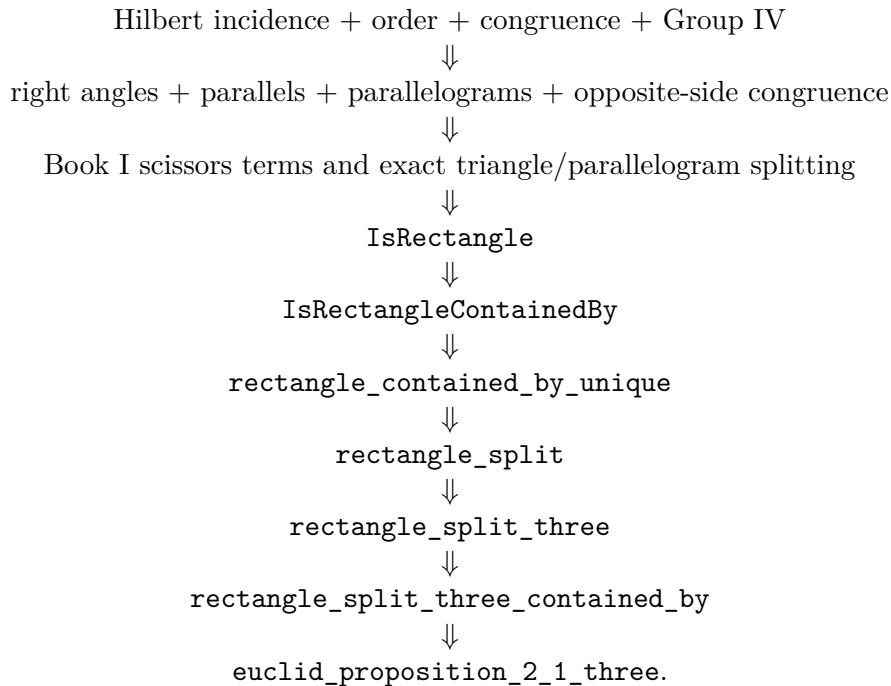
But the objects being manipulated by Euclid are still geometric figures and relations of equality between them. The documentation should therefore resist rewriting the theorem as if Euclid had stated

$$a(b + c + d) = ab + ac + ad$$

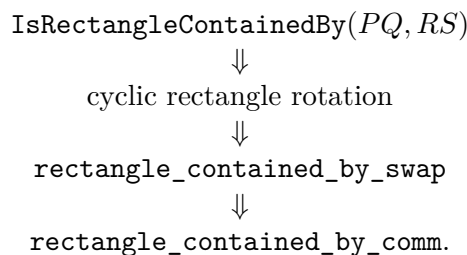
in symbolic algebra. That equation is a modern shadow of the configuration, not the proposition itself.

1.16 Dependency Summary

The actual formal dependency chain is



The symmetric side-order branch is



New geometric axiom in II.1:	no
New proposition-local axiom:	no
New rectangle abstraction:	yes, in <code>HilbertRectangle</code>
New reusable split helper:	yes
New representative-independence theorem:	yes
Exact scissors dissection used:	yes
General equicomplementability required by II.1:	no
Numerical area semantics:	no
Segment multiplication / field theory:	no
Group IV available/used transitively:	yes
Continuity assumption in the file:	no
Proposition-local <code>sorry</code> /admit:	no
Modular build:	clean

1.17 Conclusion

Proposition II.1 is the first clean instance in Book II of a product-like law being realized entirely by geometry. A segment is divided; a rectangle is cut along corresponding parallels; exact scissors equality identifies the whole with the sum of the pieces. The rectangle layer then removes the accidental choice of a concrete drawing, so Euclid’s phrase “the rectangle contained by” becomes a representative-independent synthetic magnitude.

The formalization also isolates the two structural laws that make the language look algebraic. Additivity comes from actual dissection. Commutativity comes from cyclic rotation of the rectangle. A square is the special case in which the two containing segments agree. None of these steps requires a product of segment lengths or a numerical area map.

Thus the “geometric algebra” of II.1 is best understood not as hidden modern symbolism but as a calculus of geometric magnitudes whose algebraic laws emerge from constructions, congruence, and finite decomposition.

The representative-independent three-part theorem at the end of the current formalization is:

```

theorem euclid_proposition_2_1_three
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]

  -- Concrete dissection diagram.
  (B C D E L M K N X Y P Q : Geo.Point)
  (hRect : IsRectangle Geo B C E D)
  (hCE_PQ : Geo.Congruent C E P Q)

  -- First cut.
  (hBMC : Geo.Between B M C)
  (hDLE : Geo.Between D L E)
  (hDXC : Geo.Between D X C)
  (hMXL : Geo.Between M X L)
  (hLeftPar :
    IsParallelogram Geo L D B M)
  (hRightPar :
    IsParallelogram Geo C E L M)

  -- Second cut.
  (hMNC : Geo.Between M N C)

```



```

(hLKE : Geo.Between L K E)
(hLYC : Geo.Between L Y C)
(hNYK : Geo.Between N Y K)
(hMiddlePar :
  IsParallelogram Geo K L M N)
(hFinalPar :
  IsParallelogram Geo C E K N)

-- Arbitrary representatives of the four rectangles.
(U0 U1 U2 U3 : Geo.Point)
(V0 V1 V2 V3 : Geo.Point)
(W0 W1 W2 W3 : Geo.Point)
(Z0 Z1 Z2 Z3 : Geo.Point)

(hWhole :
  IsRectangleContainedBy Geo
    U0 U1 U2 U3 B C P Q)

(hPart1 :
  IsRectangleContainedBy Geo
    V0 V1 V2 V3 B M P Q)

(hPart2 :
  IsRectangleContainedBy Geo
    W0 W1 W2 W3 M N P Q)

(hPart3 :
  IsRectangleContainedBy Geo
    Z0 Z1 Z2 Z3 N C P Q) :
HilbertScissorsEq Geo
  (hilbertParallelogramTerm Geo U0 U1 U2 U3)
  (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
    (hilbertParallelogramTerm Geo W0 W1 W2 W3 +
      hilbertParallelogramTerm Geo Z0 Z1 Z2 Z3)) := by

```

Chapter 2

Proposition II.2

2.1 Introduction

Proposition II.2 is the first place where the rectangle calculus established in II.1 is reused rather than developed further. The classical proposition is visually elementary: a square on a straight line is cut by a parallel through an arbitrary point of that line, and the two resulting rectangles fill the square. The formal reconstruction preserves exactly that geometric content, but its proof architecture is deliberately different from Euclid’s.

Euclid proves II.2 directly from a square construction and a parallel through the point of division. The current Lean theorem instead imports `Proposition2_1` and applies the representative-independent binary form of II.1. One additional operation is needed: Euclid names each part rectangle with the whole side first, whereas the II.1 cut theorem returns the divided part first. The proof therefore rotates each rectangle representative by one vertex, applies II.1, and then transports the result back to Euclid’s original order.

This makes II.2 a particularly clean test of the Book II API. Two algebra-like laws are already available, but neither is primitive algebra:

- additivity is realized by an exact rectangle dissection;
- exchange of the two containing segments is realized by cyclic rotation of a rectangle.

No multiplication of segment lengths and no numerical area function occurs in the theorem.

2.2 Euclid’s Statement and Classical Configuration

2.2.1 The statement

In Heath’s wording, Proposition II.2 states:

If a straight line be cut at random, the rectangle contained by the whole and both of the segments is equal to the square on the whole.

Here “both of the segments” means the two rectangles obtained by pairing the whole line with each of the two parts. If AB is cut at C , the geometric statement is therefore

$$\text{Rect}(AB, AC) + \text{Rect}(AB, CB) = \text{Sq}(AB).$$

The notation is documentary shorthand only. It does not introduce a number-valued rectangle function.

2.2.2 The classical diagram

Euclid takes a straight line AB cut at C , constructs the square $ADEB$ on AB by I.46, and through C draws CF parallel to the two opposite sides AD and BE by I.31 [1]. The square is thereby partitioned into two rectangles. One is contained by the whole AB and the segment AC ; the other is contained by the whole AB and the segment CB .

The current Lean configuration uses different letters:

classical role	Lean role
AB	BC
C	M
$ADEB$	$BCED$
CF	ML

Thus $B - M - C$, the outer figure $BCED$ is a square, and the cut ML divides it into the left and right rectangular pieces.

Figure 2.1 shows both the configured cut and the separate representation change used by the formal proof.

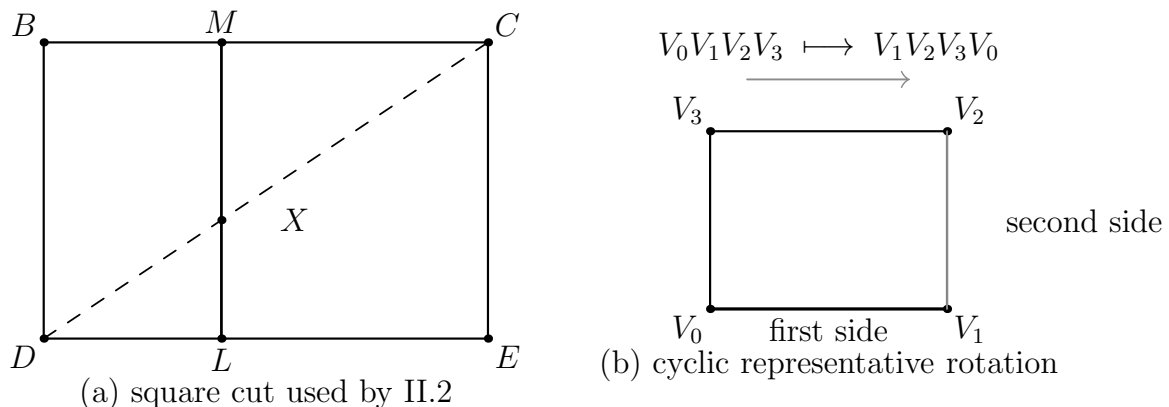


Figure 2.1: Two mechanisms used in the formal II.2 proof. Panel (a) is the instantiated square $BCED$ with $B - M - C$ and the cut ML ; the dashed diagonal DC meets the cut at the hidden witness X , so the formal data include both $D - X - C$ and $M - X - L$. Panel (b) shows the cyclic presentation change $V_0V_1V_2V_3 \mapsto V_1V_2V_3V_0$, which exchanges the two adjacent side roles while preserving exact scissors content.

2.3 Classical Proof

The classical proof has only four mathematical steps.

First, construct the square on the whole segment. Second, draw through the point of section a line parallel to the appropriate pair of sides of the square. Third, observe that this line partitions the square into two rectangles. Finally, identify those rectangles by Book II Definition 1: their adjacent sides are the whole segment together with the two parts of the divided segment.

In the standard lettering this gives

$$\text{Rect}(AB, AC) + \text{Rect}(AB, CB) = \text{Sq}(AB).$$

Euclid does not invoke II.1 in this proof. The local classical dependency graph is instead

$$\begin{array}{c} \text{I.46: construct the square on } AB \\ \Downarrow \\ \text{I.31: draw the parallel through } C \\ \Downarrow \\ \text{II.Def.1: identify the two rectangles} \\ \Downarrow \\ \text{the two pieces constitute the square.} \end{array}$$

This fact is important for documentation. The formal theorem proves the same geometric identity, but it is not a line-by-line formalization of Euclid’s historical derivation.

2.4 The Geometric Identity

In the current Lean lettering the square side is BC and the cut point is M . The two Euclidean rectangles are therefore

$$\text{Rect}(BC, BM), \quad \text{Rect}(BC, MC),$$

and the theorem asserts the exact scissors relation

$$\text{Rect}(BC, BM) + \text{Rect}(BC, MC) \simeq_{\text{sc}} \text{Sq}(BC).$$

A modern algebraic shadow is obtained by writing $x = y + z$:

$$x^2 = xy + xz.$$

This is useful as a mnemonic for the pattern, but it is not the object proved by Lean and it is not a claim that Euclid has defined multiplication on segment lengths. The formal terms remain rectangle representatives and their exact scissors relation.

2.5 Reusable Book II Infrastructure

II.2 imports

```
import CGJteamLab.Proposition2_1
```

and introduces no new rectangle abstraction. Its proof consumes four pieces of the existing Book II infrastructure.

2.5.1 A square as a rectangle contained by the same side twice

From

```
hSquare : IsSquare Geo B C E D
```

the theorem

```
square_is_rectangle_contained_by_side
```

produces

```
IsRectangleContainedBy Geo B C E D B C B C
```

so the concrete square is recognized synthetically as a rectangle contained by BC, BC .

2.5.2 The binary form of II.1

The reusable distributive kernel is

```
euclid_proposition_2_1_two
```

whose geometric content is

$$\text{Rect}(BC, PQ) \simeq_{\text{sc}} \text{Rect}(BM, PQ) + \text{Rect}(MC, PQ)$$

when $B - M - C$ and the corresponding rectangular cut data are supplied.

For II.2 the fixed segment is specialized to $PQ = BC$. Thus the right-hand pieces naturally arrive in the order

$$\text{Rect}(BM, BC), \quad \text{Rect}(MC, BC).$$

2.5.3 Exchanging the containing-side order

Euclid’s statement uses the opposite order:

$$\text{Rect}(BC, BM), \quad \text{Rect}(BC, MC).$$

The current proof does not treat this as “commutativity of multiplication.” It calls

```
rectangle_contained_by_swap
```

on each arbitrary representative.

The theorem rotates a rectangle

$$A B C D \mapsto B C D A,$$

returns the corresponding `IsRectangleContainedBy` fact with the two containing segments exchanged, and simultaneously returns a `HilbertScissorsEq` relating the two cyclic presentations.

The higher-level theorem `rectangle_contained_by_comm` belongs to the same rectangle API, but the public II.2 proof uses `rectangle_contained_by_swap` directly because its returned data are exactly the swapped representatives and exact scissors transports needed for the subsequent II.1 call.

2.6 Proof Architecture of the Reconstruction

The Lean proof has four stages.

2.6.1 Step 1: recognize the square

The square hypothesis is converted to the “contained by” language:

```
have hSquareContained :
  IsRectangleContainedBy Geo
    B C E D B C B C :=
square_is_rectangle_contained_by_side
  Geo B C E D hSquare
```

The rectangle structure and the side congruence $CE \cong BC$ are then extracted from this fact.

2.6.2 Step 2: rotate the two arbitrary part representatives

The two hypotheses supplied in Euclid’s order are

```
hPart1 :
  IsRectangleContainedBy Geo
    V0 V1 V2 V3 B C B M

hPart2 :
  IsRectangleContainedBy Geo
    W0 W1 W2 W3 B C M C
```

The current source then applies `rectangle_contained_by_swap` separately to `hPart1` and `hPart2`. Each call returns a swapped `IsRectangleContainedBy` fact and an exact scissors equality for the cyclically rotated representative. The resulting representatives are $V_1V_2V_3V_0$ and $W_1W_2W_3W_0$.

The mathematical result is that the rotated representatives now have the side orders required by the II.1 binary theorem.

2.6.3 Step 3: apply II.1 with fixed segment BC

The central call is

```
have hSplit :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo B C E D)
    (hilbertParallelogramTerm Geo V1 V2 V3 V0 +
      hilbertParallelogramTerm Geo W1 W2 W3 W0) :=
euclid_proposition_2_1_two
  Geo
  B C D E L M X
  B C
  hRect
```

```

hCE_BC
hBMC
hDLE
hDXC
hMXL
hLeftPar
hRightPar
B C E D
V1 V2 V3 V0
W1 W2 W3 W0
hSquareContained
hPart1Swap
hPart2Swap

```

This is the whole mathematical core of the reconstruction:

$$\text{Sq}(BC) \simeq_{\text{sc}} \text{Rect}(BM, BC) + \text{Rect}(MC, BC).$$

2.6.4 Step 4: return to Euclid's ordering

The two exact rotation equalities are added:

```

have hPartsBack :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo V1 V2 V3 V0 +
     hilbertParallelogramTerm Geo W1 W2 W3 W0)
    (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
     hilbertParallelogramTerm Geo W0 W1 W2 W3) :=
  HilbertScissorsEq.add
    (Geo := Geo)
    (HilbertScissorsEq.symm
     (Geo := Geo) hRotate1)
    (HilbertScissorsEq.symm
     (Geo := Geo) hRotate2)

```

Transitivity composes `hSplit` with `hPartsBack`, and the final symmetry reverses the result into the order used by the public theorem: sum of the two Euclidean rectangles on the left, square on the right.

2.7 Lean Formalization

The current public theorem is the following instantiated diagram theorem:

```

theorem euclid_proposition_2_2
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]

  (B C D E L M X : Geo.Point)

  (hSquare : IsSquare Geo B C E D)

  (hBMC : Geo.Between B M C)

```

```

(hDLE : Geo.Between D L E)
(hDXC : Geo.Between D X C)
(hMXL : Geo.Between M X L)
(hLeftPar :
  IsParallelogram Geo L D B M)
(hRightPar :
  IsParallelogram Geo C E L M)

(V0 V1 V2 V3 : Geo.Point)
(W0 W1 W2 W3 : Geo.Point)

(hPart1 :
  IsRectangleContainedBy Geo
    V0 V1 V2 V3 B C B M)

(hPart2 :
  IsRectangleContainedBy Geo
    W0 W1 W2 W3 B C M C) :
HilbertScissorsEq Geo
  (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
   hilbertParallelogramTerm Geo W0 W1 W2 W3)
  (hilbertParallelogramTerm Geo B C E D) := by

```

Three aspects of the signature are mathematically significant.

First, the theorem assumes an instantiated square and cut diagram. It is not an existence theorem constructing the square and the parallel cut from only a segment and a cut point. Euclid performs those constructions in the classical proof; the present public Lean theorem starts after they have been supplied.

Second, the two part rectangles are arbitrary representatives satisfying `IsRectangleContainedBy`. Their exact placement in the plane is not fixed by the statement.

Third, the conclusion is the exact relation `HilbertScissorsEq`. It is not merely a conclusion in the more general `HilbertScissorsEquicomplementable` relation. II.2 remains at the exact-dissection level because its proof is ultimately the binary exact split of II.1 plus exact representation transports.

2.8 Relationship with Proposition II.1

The contrast between the classical and formal dependency graphs is especially clear here.

Euclid's II.2 is proved directly from the square and the parallel cut. The current Lean file instead begins with

```
import CGJteamLab.Proposition2_1
```

and the public proof calls

```
euclid_proposition_2_1_two
```

once.

Thus II.2 is an explicit formal corollary of the binary II.1 API. No new dissection lemma is proved. The only additional mathematical work is conversion between the two possible orders of the containing segments.

This reuse also explains why the symmetric rectangle interface added at II.1 is not cosmetic. Without the side-order transport, the II.1 theorem would produce perfectly correct rectangles whose formal orientation does not match the language of II.2.

2.9 Relationship with Book I Infrastructure

The direct import is II.1, not a Book I proposition. Nevertheless the inherited chain underneath II.1 contains the Book I geometry on which the rectangle calculus rests.

The exact binary split is built over the scissors decomposition infrastructure developed around I.47. Rectangle structure uses Euclidean parallelogram theory and synthetic right angles; the square hypothesis uses the square infrastructure established in Book I. Consequently the formal proof of II.2 does not replay I.31 and I.46 as Euclid does. Those classical constructions have been absorbed into the established geometric API and into the hypotheses of the instantiated theorem.

2.10 Formalization Notes and Hidden Geometric Data

The visible square cut hides several obligations that are explicit parameters of the current theorem:

- $B - M - C$, fixing the orientation of the cut on the square side;
- $D - L - E$, locating the opposite endpoint of the cut;
- $D - X - C$ and $M - X - L$, recording the intersection X of the cut with the diagonal used by the inherited exact scissors split;
- `IsParallelogram Geo L D B M` for the left piece;
- `IsParallelogram Geo C E L M` for the right piece;
- two arbitrary representatives of the Euclidean part rectangles;
- two cyclic representative changes, each accompanied by an exact scissors equality.

The point X has no role in the elementary verbal statement. It belongs to the current exact-dissection interface inherited from the triangulated scissors proof. It is therefore documented as hidden formal geometry rather than presented as part of Euclid's original construction.

There is also an important negative observation. No `SameRay`, `SameSide`, or `OppositeSide` obligation appears in `Proposition2_2.lean` itself. The corresponding low-level geometry has already been discharged inside the reusable rectangle and II.1 theorems.

2.11 Geometry, Content, Representation, and Later Arithmetic

2.11.1 Geometry

The geometric layer consists of a square, betweenness on two opposite sides, the diagonal/cut witness, and two parallelogram pieces. Congruence of the square side with itself identifies the square as a rectangle contained by BC, BC .

2.11.2 Content / scissors

The conclusion and every transport used in the proof are exact `HilbertScissorsEq` relations. The proof uses symmetry, transitivity, and addition of exact scissors equalities. It does not weaken to the more general equicomplementability relation.

2.11.3 Representation

Representation is the distinctive issue in II.2. The theorem accepts arbitrary rectangles for

$$\text{Rect}(BC, BM) \quad \text{and} \quad \text{Rect}(BC, MC).$$

The binary II.1 theorem wants the factors in the reverse order. Cyclic rotation of each representative supplies the required conversion while preserving exact scissors content.

2.11.4 Later arithmetic / numerical area

Neither layer is used. In particular, the proof does not define a segment product, does not invoke Hilbert’s later segment field, and does not assign real areas to the square or rectangles.

2.12 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and therefore works in the same Euclidean project layer as II.1. No new proposition-specific geometric axiom or content axiom is declared in `Proposition2_2.lean`. Inspection of the current 162-line source shows one theorem declaration and no proposition-local `axiom`, `sorry`, or `admit`.

This should not be described as “axiom-free.” The theorem inherits the axioms and typeclass assumptions of the established Hilbert/Euclidean/scissors library transitively. No current `#print axioms` output was supplied with the source used for this documentation pass, so the transitive axiom list is not reproduced here.

The implementation checkpoint records that the II.2 Lean build had already been checked during theorem development. This documentation workspace does not contain the complete Lean project, so the Lean build is not re-run here; the LaTeX and Asymptote checks are reported separately.

2.13 Hartshorne’s Semantic Control

Hartshorne places the end of Book I and all of Book II inside Euclid’s theory of area, interpreted synthetically as equal content rather than as an unexplained numerical area function [5]. He also describes Book II as part of an “algebra of areas”: rectangles can play a product-like role because Euclid has no operation that multiplies two line segments to produce another segment.

II.2 fits that interpretation exactly, provided the levels are kept separate. The square on the whole and the two rectangles are geometric figures; their equality is a content statement; the modern identity $x^2 = xy + xz$ is only an interpretive shadow.

The current Lean theorem is locally stronger than the weakest equal-content reading because its configured proof stays at exact scissors equality. That strength comes from the explicit binary dissection inherited from II.1, not from a general principle identifying every equal-content pair with an exact dissection.

2.14 Hilbert’s Layering

Hilbert supplies a second semantic guardrail. His theory separates synthetic plane content from the later arithmetic of segments. Segment multiplication is introduced only in a later arithmetic construction, with a fixed unit segment and parallel geometry; it is not primitive rectangle notation.

The II.2 proof remains entirely on the earlier side of this boundary:

square and rectangle geometry
 $\Downarrow$
 exact scissors dissection
 $\Downarrow$
 representative rotation and transport.

The apparent commutativity required by the formula $xy = yx$ is realized geometrically by `rectangle_contained_by_swap`, not imported from Hilbert’s later segment arithmetic.

2.15 Dependency Summary

The actual Lean dependency chain is

Hilbert incidence + Euclidean plane layer
 $\Downarrow$
 Book I right-angle, square, parallelogram, and scissors infrastructure
 $\Downarrow$
`HilbertRectangle : IsRectangleContainedBy, rectangle_contained_by_swap`
 $\Downarrow$
 Proposition II.1 binary API
 $\Downarrow$
`euclid_proposition_2_1_two`
 $\Downarrow$
 two representative rotations + exact scissors addition/transitivity
 $\Downarrow$
`euclid_proposition_2_2.`

The classical local chain is different:

$$\text{I.46} \longrightarrow \text{I.31} \longrightarrow \text{square cut into two rectangles} \longrightarrow \text{II.2.}$$

New geometric axiom in II.2:	no
New content axiom in II.2:	no
New proposition-local helper theorem:	no
New reusable helper theorem:	no
New <code>HilbertRectangle</code> theorem introduced by II.2:	no
New Interface / Book Zero / Scissors theorem:	no
Reuses an earlier Book II proposition:	yes, II.1
Direct side-order transport used:	yes, <code>rectangle_contained_by_swap</code>
Exact <code>HilbertScissorsEq</code> :	yes
General equicomplementability required:	no
Numerical area:	no
Segment multiplication:	no
Public Lean theorem build:	previously checked; not re-run here
Full relevant Lean build:	previously checked; not re-run here
Proposition-local <code>sorry/admit</code> :	no

2.16 Conclusion

Proposition II.2 is mathematically a square cut into two rectangles, but formally it does more than repeat the picture. It demonstrates that the II.1 rectangle calculus is already reusable: the binary dissection theorem supplies additivity, and cyclic rectangle rotation reconciles the two possible orders of the containing sides.

This is also the first sharp divergence between Euclid’s local proof graph and the formal Book II graph. Euclid proves II.2 directly from I.46 and I.31. Lean uses II.1 as a library theorem and turns II.2 into a short transport-and-reuse argument. The theorem is therefore not a literal transcription of the classical proof, but it remains entirely synthetic and geometric.

The result can be remembered by the modern shadow

$$x^2 = xy + xz, \quad x = y + z,$$

but the formal content is more precise: arbitrary representatives of the two rectangles contained by the whole side and the two parts are exactly scissors-equal, in sum, to the concrete square on the whole side. No numerical area and no multiplication of segment lengths is required.

Chapter 3

Proposition II.3

3.1 Introduction

Proposition II.3 is the second immediate reuse of the rectangle calculus established in Proposition II.1. Its classical form is again a statement about a straight line cut at an arbitrary point, but now one of the two resulting segments plays a distinguished role. The rectangle contained by the whole line and that segment is decomposed into a rectangle contained by the two parts and the square on the distinguished segment.

The current Lean reconstruction makes this structure unusually transparent. Its only direct import is the II.1 module:

```
import CGJteamLab.Proposition2_1
```

The proof first recognizes the supplied square on MC as a rectangle contained by MC, MC , and then applies the binary form of II.1 with the fixed segment specialized to MC . No new dissection theorem, side-order swap, or proposition-local helper theorem is needed.

This is an important contrast with II.2. There the Euclidean order of the two containing segments did not match the order produced by the binary II.1 theorem, so cyclic rectangle rotation was required. In II.3 the order already matches:

$$\text{Rect}(BC, MC) \simeq_{\text{sc}} \text{Rect}(BM, MC) + \text{Rect}(MC, MC).$$

The square on MC is then simply a chosen representative of the final $\text{Rect}(MC, MC)$ term.

Thus II.3 is not formalized through multiplication of segment lengths or through a numerical area function. Its apparently algebraic shape is realized by an exact geometric cut together with representative independence in the rectangle API.

3.2 Euclid's Statement and Classical Configuration

3.2.1 The statement

In Heath's wording, Proposition II.3 states:

If a straight line be cut at random, the rectangle contained by the whole and one of

the segments is equal to the rectangle contained by the segments and the square on the aforesaid segment.

If AB is cut at C and the distinguished part is BC , the geometric identity is

$$\text{Rect}(AB, BC) = \text{Rect}(AC, CB) + \text{Sq}(BC).$$

As throughout the Book II documentation, Rect and Sq are expository shorthand for geometric figures and their content. They are not number-valued functions.

3.2.2 The classical diagram

Euclid constructs the square $CDEB$ on the distinguished segment CB by Proposition I.46. The side ED is produced, and through A a line is drawn parallel to the corresponding sides of the square by Proposition I.31 [1]. The resulting larger rectangle has adjacent side magnitudes AB and BC . The line through C separates it into a rectangle with side magnitudes AC, CB and the square on CB .

The current Lean lettering is related to the classical roles by

classical role	Lean role
AB	BC
C	M
AC	BM
CB	MC

so the formal cut relation is $B - M - C$ and the distinguished segment is MC . The configured outer rectangle is $BCED$, with $CE \cong MC$, while the cut through M meets the opposite side at L .

Figure 3.1 shows the geometry used by the reconstruction. The rectangle $BCED$ is drawn below the segment BC . Its right-hand piece has width MC and height congruent to MC , so it realizes the square-like $\text{Rect}(MC, MC)$ part of the dissection. The supplied square $MCFG$ is drawn on the opposite side of MC to emphasize a formal point: the conclusion may use a square representative different from the concrete piece created by the cut.

3.3 Classical Proof

Let AB be cut at C , and choose BC as the segment named in the proposition. Construct the square $CDEB$ on CB by I.46. Produce one side of the square and through A draw a line parallel to the opposite sides by I.31. This forms a larger rectangle whose adjacent side magnitudes are AB and BC .

The line through C partitions that larger rectangle into two pieces. The first has adjacent side magnitudes AC and CB , and the second is precisely the square on CB . Hence

$$\text{Rect}(AB, BC) = \text{Rect}(AC, CB) + \text{Sq}(BC).$$

This is Euclid's entire content argument: the whole rectangle is constituted by the two displayed subfigures.

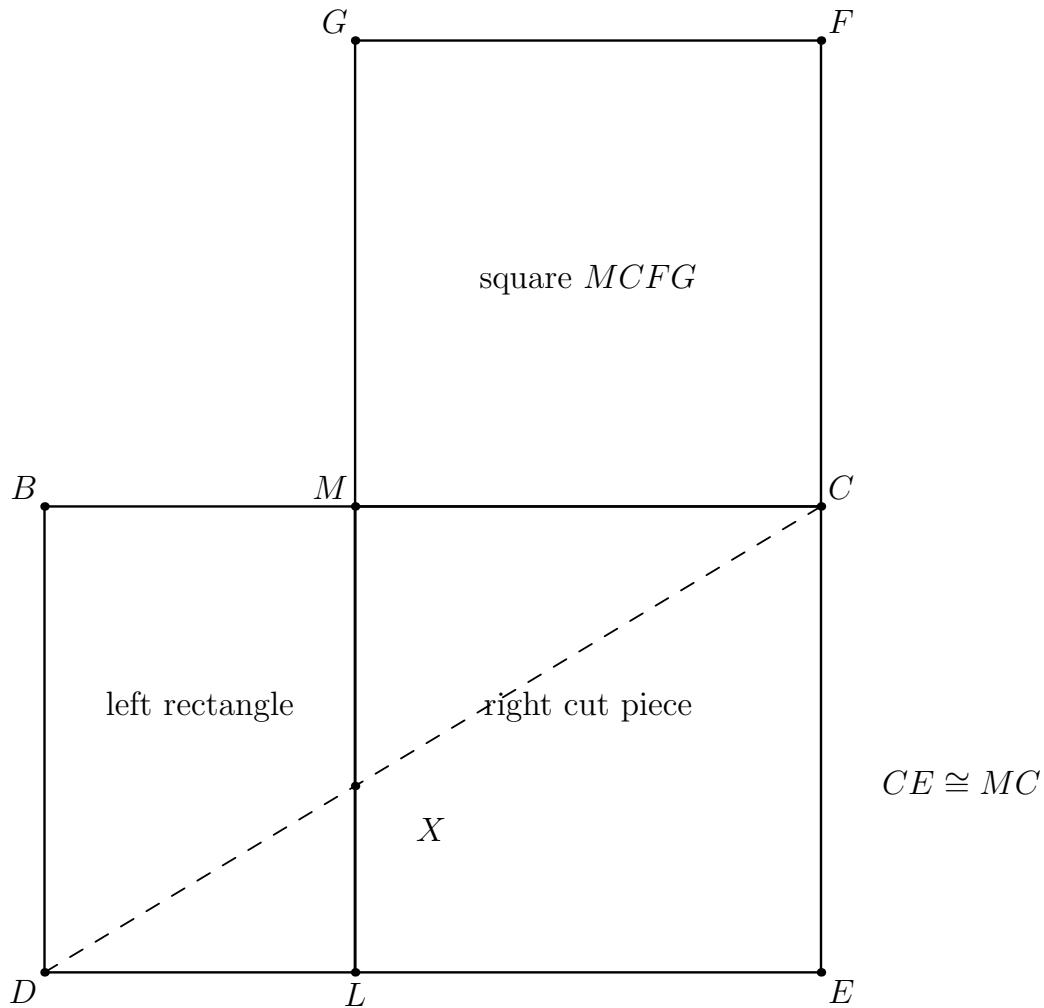


Figure 3.1: The geometric mechanism of II.3. The outer rectangle $BCED$ has height $CE \cong MC$ and is split by ML at the point $B - M - C$. The dashed diagonal DC meets the cut at the hidden witness X . The right-hand cut piece represents $\text{Rect}(MC, MC)$; the separately drawn square $MCFG$ is the square representative used in the public theorem.

The local classical dependency graph is therefore

$$\begin{array}{c}
 \text{I.46: construct the square on } CB \\
 \Downarrow \\
 \text{I.31: draw the required parallel through } A \\
 \Downarrow \\
 \text{Book II Definition 1: identify the rectangles} \\
 \Downarrow \\
 \text{whole rectangle} = \text{rectangle} + \text{square} \\
 \Downarrow \\
 \text{II.3.}
 \end{array}$$

Euclid does not cite Proposition II.1 in this proof. The current Lean proof, by contrast, is explicitly organized as a corollary of the reusable II.1 binary theorem. The classical and formal dependency graphs therefore prove the same geometric identity through different library organizations.

3.4 The Geometric Identity

In the current Lean lettering, $B - M - C$. The theorem fixes the segment MC and applies it to the whole BC and the two parts BM, MC . Its exact geometric content is

$$\text{Rect}(BC, MC) \simeq_{\text{sc}} \text{Rect}(BM, MC) + \text{Sq}(MC).$$

A modern algebraic shadow is obtained by writing

$$x = y + z,$$

where x corresponds to BC , y to BM , and z to MC . Then the pattern is

$$xz = yz + z^2,$$

or equivalently

$$(y + z)z = yz + z^2.$$

This formula is useful only as a mnemonic. The Lean proof never forms the products xz or yz as segment-arithmetic objects. The actual objects are rectangle representatives and a square, related by exact scissors equality.

3.5 Reusable Book II Infrastructure

The current file begins with

```
import CGJteamLab.Proposition2_1
```

and introduces no new rectangle abstraction.

3.5.1 The binary form of II.1

The reusable theorem is

```
euclid_proposition_2_1_two
```

with geometric content

$$\text{Rect}(BC, PQ) \simeq_{\text{sc}} \text{Rect}(BM, PQ) + \text{Rect}(MC, PQ)$$

whenever $B - M - C$ and the required rectangular cut data are supplied.

For II.3 the fixed segment is specialized directly to

$$PQ = MC.$$

Therefore II.1 returns

$$\text{Rect}(BC, MC) \simeq_{\text{sc}} \text{Rect}(BM, MC) + \text{Rect}(MC, MC),$$

which already has exactly the containing-side order required by the public II.3 theorem.

3.5.2 A square as $\text{Rect}(MC, MC)$

The square hypothesis

```
hSquare : IsSquare Geo M C F G
```

is converted by

```
square_is_rectangle_contained_by_side
```

into

```
IsRectangleContainedBy Geo M C F G M C M C
```

so the supplied square is a concrete representative of the final $\text{Rect}(MC, MC)$ term.

3.5.3 What II.3 does not use

Unlike II.2, the current proof does not call

```
rectangle_contained_by_swap  
rectangle_contained_by_comm
```

The absence is mathematically informative. The binary II.1 theorem already places BM and MC first and the fixed segment MC second, exactly as II.3 needs. No geometric analogue of exchanging factors is required.

3.6 Proof Architecture of the Reconstruction

The public proof has only two mathematical stages.

3.6.1 Step 1: recognize the square

The first local fact is

```
have hSquareContained :  
  IsRectangleContainedBy Geo  
    M C F G M C M C :=  
  square_is_rectangle_contained_by_side  
    Geo M C F G hSquare
```

This moves from the square-specific geometric predicate to the general “rectangle contained by” language used by II.1.

3.6.2 Step 2: instantiate II.1 with the fixed segment MC

The remainder of the proof is one direct call:

```
exact
  euclid_proposition_2_1_two
    Geo
    B C D E L M X
    M C
    hRect
    hCE_MC
    hBMC
    hDLE
    hDXC
    hMXL
    hLeftPar
    hRightPar
    U0 U1 U2 U3
    V0 V1 V2 V3
    M C F G
    hWhole
    hProduct
    hSquareContained
```

No scissors transitivity, addition, symmetry, or representative rotation is performed locally in II.3. Those mechanisms have already been packaged inside the representative-independent binary theorem of II.1.

This makes the formal DAG exceptionally short:

$$\begin{array}{c}
 \text{square } MCFG \\
 \Downarrow \\
 \text{square_is_rectangle_contained_by_side} \\
 \Downarrow \\
 \text{Rect}(MC, MC) \text{ representative} \\
 \Downarrow \\
 \text{euclid_proposition_2_1_two with } PQ = MC \\
 \Downarrow \\
 \text{euclid_proposition_2_3.}
 \end{array}$$

3.7 Lean Formalization

The current public theorem is:

```
theorem euclid_proposition_2_3
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]

  (B C D E L M X : Geo.Point)
  (hRect : IsRectangle Geo B C E D)
  (hCE_MC : Geo.Congruent C E M C)

  (hBMC : Geo.Between B M C)
```

```

(hDLE : Geo.Between D L E)
(hDXC : Geo.Between D X C)
(hMXL : Geo.Between M X L)
(hLeftPar :
  IsParallelogram Geo L D B M)
(hRightPar :
  IsParallelogram Geo C E L M)

(F G : Geo.Point)
(hSquare : IsSquare Geo M C F G)

(U0 U1 U2 U3 : Geo.Point)
(V0 V1 V2 V3 : Geo.Point)

(hWhole :
  IsRectangleContainedBy Geo
    U0 U1 U2 U3 B C M C)

(hProduct :
  IsRectangleContainedBy Geo
    V0 V1 V2 V3 B M M C) :
HilbertScissorsEq Geo
  (hilbertParallelogramTerm Geo U0 U1 U2 U3)
  (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
    hilbertParallelogramTerm Geo M C F G) := by

```

Several features of this signature are mathematically significant.

First, the theorem assumes a configured outer rectangle and its cut. It does not construct that diagram from the bare data $B - M - C$. Euclid performs the square and parallel constructions in his proof; the formal theorem receives the corresponding geometry as explicit hypotheses.

Second, $U_0U_1U_2U_3$ and $V_0V_1V_2V_3$ are arbitrary representatives of

$$\text{Rect}(BC, MC) \quad \text{and} \quad \text{Rect}(BM, MC).$$

The conclusion is therefore not restricted to the literal pieces drawn in the configured cut.

Third, the square $MCFG$ is concrete, but the II.1 representative-independence machinery permits it to stand for the $\text{Rect}(MC, MC)$ piece even when it is not the internal right-hand rectangle of the supplied cut diagram.

Fourth, the conclusion is exactly

```
HilbertScissorsEq
```

rather than the more general relation

```
HilbertScissorsEquicomplementtable
```

and it is not equality of real numbers.

3.8 Relationship with Earlier Book II Propositions

3.8.1 Proposition II.1

II.3 is a direct formal corollary of II.1. The import is

```
import CGJteamLab.Proposition2_1
```

and the only substantial theorem called in the final proof is

```
euclid_proposition_2_1_two
```

The specialization $PQ = MC$ converts the general binary distributive pattern into II.3 immediately.

The relationship can be written schematically as

$$\text{Rect}(BC, PQ) = \text{Rect}(BM, PQ) + \text{Rect}(MC, PQ)$$

followed by

$$PQ := MC.$$

The final term then becomes $\text{Rect}(MC, MC)$, represented by the square on MC .

3.8.2 Proposition II.2

The current Lean file does *not* import Proposition II.2, and its public theorem is not used in the proof. II.2 and II.3 are therefore sibling corollaries of the II.1 binary API rather than a formal chain $\text{II.1} \rightarrow \text{II.2} \rightarrow \text{II.3}$.

The contrast is nevertheless useful. II.2 required cyclic representative rotation because its chosen side order was opposite to the order returned by II.1. II.3 chooses the fixed segment MC in precisely the position expected by II.1, so even that transport disappears. This is evidence that the II.1 API is not merely sufficient but well aligned with the geometry of the next Book II identities.

3.9 Relationship with Book I Infrastructure

Euclid's local proof explicitly calls I.46 to construct the square and I.31 to draw the parallel. The current Lean theorem does not replay those constructions. Instead it assumes the resulting rectangle, cut, parallelograms, side congruence, and square as data.

Below II.1, however, the formal dependency chain still rests on the Book I infrastructure for rectangles, right angles, squares, parallelograms, and exact scissors decomposition. The binary split ultimately uses the exact rectangle cut machinery developed over the Book I scissors layer. Thus the difference is one of modular organization, not a change from synthetic to numerical geometry.

3.10 Formalization Notes and Hidden Geometric Data

The elementary picture hides several obligations that appear explicitly in the current theorem signature:

- $B - M - C$, fixing the cut point and its orientation on the whole side;
- $D - L - E$, locating the opposite endpoint of the cut;
- $D - X - C$ and $M - X - L$, recording the intersection X of the diagonal and the cut line used by the inherited exact scissors theorem;
- `IsParallelogram Geo L D B M` for the left piece;
- `IsParallelogram Geo C E L M` for the right piece;
- $CE \cong MC$, making the outer rectangle have the required fixed second side magnitude;
- a supplied square $MCFG$;
- arbitrary representatives of the whole rectangle and the BM -by- MC product rectangle.

The point X is invisible in Euclid’s verbal statement. It belongs to the triangulated exact-dissection interface inherited from II.1. It should therefore be read as hidden formal geometry rather than as an additional step in Euclid’s classical construction.

There are also useful negative observations. No `SameRay`, `SameSide`, or `OppositeSide` hypothesis occurs in `Proposition2_3.lean` itself, and no side-order swap is performed. Those lower-level obligations have already been absorbed by the reusable rectangle and II.1 layers.

3.11 Geometry, Content, Representation, and Later Arithmetic

3.11.1 Geometry

The geometric layer consists of the rectangle $BCED$, the cut point M , the opposite cut point L , the diagonal witness X , the two parallelogram pieces, the congruence $CE \cong MC$, and the square $MCFG$.

3.11.2 Content / scissors

The conclusion is the exact relation `HilbertScissorsEq`. The public proof does not weaken the result to general equicomplementability. Exactness is inherited from the binary rectangle split and representative transports already proved in II.1.

3.11.3 Representation

Representation is present but simpler than in II.2. The theorem accepts an arbitrary rectangle representative for $\text{Rect}(BC, MC)$ and an arbitrary representative for $\text{Rect}(BM, MC)$, together with a concrete square on MC . The binary II.1 theorem is already independent of the particular concrete rectangles appearing in the cut, so no additional representation transport is required locally.

3.11.4 Later arithmetic / numerical area

Neither layer is used. No segment product is defined, no fixed unit segment is introduced, and no real-valued area function appears. The expression $xz = yz + z^2$ is only the modern algebraic shadow of a geometric scissors identity.

3.12 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and therefore lives in the same Euclidean project layer as II.1 and II.2. Inspection of the current 96-line source shows one public theorem declaration and no proposition-local `axiom`, `sorry`, or `admit`.

This should not be described as “axiom-free.” The result inherits the axioms and typeclass assumptions of the established Hilbert/Euclidean/scissors library transitively. No current `#print axioms` output for II.3 was supplied with the documentation sources, so a transitive axiom list is not asserted here.

The implementation checkpoint records that the II.3 Lean theorem had already been build-checked during formalization. The present documentation pass uses the current source as authoritative but does not contain the full Lean project needed to repeat that build locally.

3.13 Hartshorne’s Semantic Control

Hartshorne’s general analysis of Euclid’s area theory states that the results of Book II are valid when Euclid’s equality of figures is read as equality of content [5]. His condensed list of selected Book II propositions does not separately single out II.3, so the source should not be cited as providing a special proof of this proposition.

The general semantic point is sufficient. II.3 compares a rectangle with the union of another rectangle and a square. In the present reconstruction this comparison is sharpened, for this configured case, to exact scissors equality. The modern identity

$$xz = yz + z^2$$

therefore belongs to Hartshorne’s “algebra of areas” only as an interpretive shadow of operations on geometric figures.

II.3 also clarifies the representative issue. The term called “the square on MC ” need not be the literal internal square-shaped piece of the cut; the formal theorem accepts the separately supplied square $MCFG$ because the rectangle calculus already proves independence of representatives at the scissors level.

3.14 Hilbert's Layering

Hilbert provides the complementary warning against reading later segment arithmetic into the proof. His synthetic theory of plane content distinguishes finite equidecomposition from the more general content relations, while his arithmetic of segments is introduced later and requires a fixed unit segment and additional parallel constructions [2].

II.3 remains entirely below that arithmetic boundary. Its formal chain is

$$\begin{array}{c} \text{rectangle and square geometry} \\ \Downarrow \\ \text{exact binary dissection from II.1} \\ \Downarrow \\ \text{representative-independent exact scissors equality.} \end{array}$$

In fact II.3 is even more direct than II.2: no geometric commutativity theorem is needed because the ordered containing segments already agree with the II.1 binary interface.

3.15 Dependency Summary

The actual Lean dependency chain is

$$\begin{array}{c} \text{Hilbert incidence + Euclidean plane layer} \\ \Downarrow \\ \text{Book I right-angle, square, parallelogram, and scissors infrastructure} \\ \Downarrow \\ \text{HilbertRectangle} \\ \text{IsRectangleContainedBy} \\ \text{square_is_rectangle_contained_by_side} \\ \Downarrow \\ \text{Proposition II.1 binary API} \\ \text{euclid_proposition_2_1_two} \\ \text{specialized by } PQ = MC \\ \Downarrow \\ \text{euclid_proposition_2_3.} \end{array}$$

The classical local chain is different:

$$\text{I.46} \longrightarrow \text{I.31} \longrightarrow \text{rectangle cut into rectangle + square} \longrightarrow \text{II.3.}$$

New geometric axiom in II.3:	no
New content axiom in II.3:	no
New proposition-local helper theorem:	no
New reusable helper theorem:	no
New <code>HilbertRectangle</code> theorem introduced by II.3:	no
New Interface / Book Zero / Scissors theorem:	no
Reuses an earlier Book II proposition:	yes, II.1
Imports II.2:	no
Direct side-order transport used:	no
Exact <code>HilbertScissorsEq</code> :	yes
General equicomplementability required:	no
Numerical area:	no
Segment multiplication:	no
Public Lean theorem build:	previously checked; not re-run here
Full relevant Lean build:	previously checked; not re-run here
Proposition-local <code>sorry/admit</code> :	no

3.16 Conclusion

Proposition II.3 is one of the cleanest tests of the Book II rectangle API. Classically, Euclid constructs a square on one segment and enlarges it by a parallel construction until the whole rectangle contained by the original line and that segment is obtained. Formally, the same identity is already contained in the binary theorem of II.1.

The decisive specialization is simply

$$PQ = MC.$$

It turns the second part rectangle of II.1 into $\text{Rect}(MC, MC)$, and the supplied square on MC provides exactly that representative. Nothing else is needed.

This also explains the precise relation between II.2 and II.3. Both are formal corollaries of II.1, but II.2 needs geometric side-order transport while II.3 does not. The result is therefore a very short Lean proof without being an algebraic proof in the modern numerical sense: the content remains rectangles, squares, congruent sides, and exact scissors equality throughout.

Chapter 4

Proposition II.4

4.1 Introduction

Proposition II.4 is the first proposition of Book II in the present reconstruction whose formal proof is naturally expressed as a composition of *two different earlier Book II propositions*. The classical theorem says that when a segment is cut into two parts, the square on the whole is equal to the two squares on the parts together with twice the rectangle contained by the parts.

The modern algebraic shadow is the familiar identity

$$(y + z)^2 = y^2 + z^2 + 2yz.$$

The formal proof, however, does not define a product of segment lengths and does not evaluate a numerical area. It remains entirely inside the rectangle and exact-scissors layer established at the beginning of Book II.

The current Lean source begins with

```
import CGJteamLab.Proposition2_2
import CGJteamLab.Proposition2_3
```

and the proof uses those imports exactly as the architecture suggests:

1. Proposition II.2 splits the square on the whole segment into two rectangles pairing the whole with the two parts;
2. Proposition II.3 expands each of those rectangles into one square and one cross rectangle;
3. representation transport identifies the two cross rectangles as two representatives of the same “rectangle contained by” the two parts;
4. addition and transitivity in `HilbertScissorsEq` combine the results.

Thus the substantial new fact at II.4 is not a new geometric construction or a new arithmetic law. It is the observation that the Book II rectangle API is already compositional enough to express the square-of-a-sum pattern using previously proved exact scissors theorems.

4.2 Euclid's Statement and Classical Configuration

4.2.1 The statement

In Heath's wording, Proposition II.4 states:

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.

Let AB be cut at C . In the geometric shorthand used throughout this volume, the statement is

$$\text{Sq}(AB) = \text{Sq}(AC) + \text{Sq}(CB) + 2 \text{Rect}(AC, CB).$$

Here Sq and Rect are expository names for geometric figures and their content. The symbol 2 means that two equal rectangle contents occur in the decomposition; it is not scalar multiplication inside the Lean segment language.

4.2.2 The classical diagram

Euclid describes the square $ADEB$ on AB by I.46 and joins the diagonal BD . Through the cut point C he draws CF parallel to the vertical sides of the square, and through the intersection G of CF with BD he draws HK parallel to AB and DE by I.31 [1]. The square is thereby divided into four subfigures named AG , CK , HF , and GE .

The figure is the classical Euclidean configuration. It is deliberately not presented as the literal Lean input diagram. The formal theorem receives three configured rectangle-cut diagrams inherited from II.2 and II.3, rather than reconstructing this single classical picture point by point.

4.3 Classical Proof

Euclid's proof first identifies the lower-right parallelogram $CGKB$ as a square on CB . Since $CF \parallel AD$, the diagonal BD gives the relevant angle equality by I.29. The square has $BA = AD$, so I.5 gives equal base angles in triangle ADB . Combining these angle equalities and using I.6 yields

$$BC = CG.$$

Opposite sides in the surrounding parallelograms are equal by I.34, so the four sides of $CGKB$ are equal. Right-angle information from the parallels and the square then shows that $CGKB$ is a square on CB [1].

The same reasoning identifies HF as the square on AC . The two remaining subfigures AG and GE are complements about the diagonal of the outer square, and Proposition I.43 gives

$$AG = GE.$$

Moreover AG has adjacent side magnitudes AC and CG , while $CG = CB$. Hence AG is the rectangle contained by AC, CB , and $AG + GE$ is twice that rectangle. Since the four subfigures constitute the whole square $ADEB$, Euclid obtains the required equality [1].

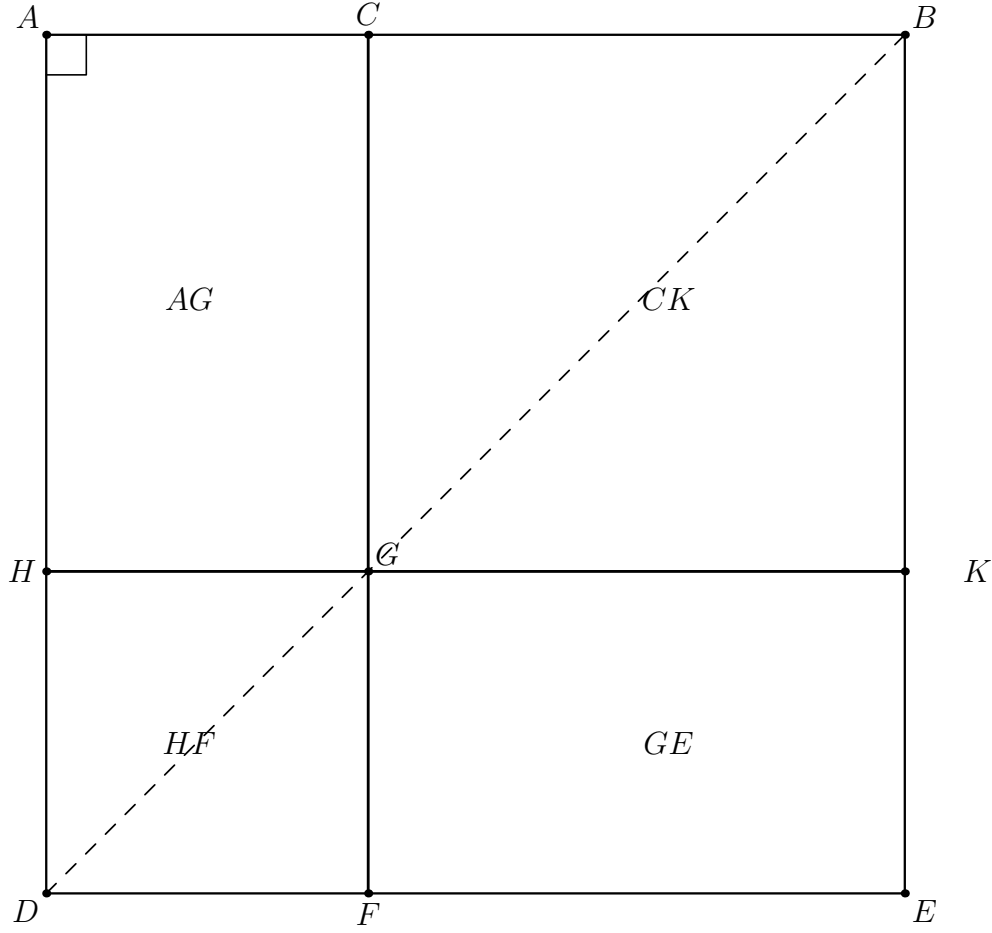


Figure 4.1: The classical configuration of Euclid II.4. The square $ADEB$ is cut at C by CF , the diagonal BD meets it at G , and the parallel HK through G produces four subfigures. Euclid proves that HF and CK are the squares on AC and CB , while the complements AG and GE are equal; AG is the rectangle contained by AC, CB .

The local classical dependency graph is therefore

$$\begin{array}{c}
 \text{I.46: square on } AB \quad + \quad \text{I.31: required parallels} \\
 \Downarrow \\
 \text{I.29} + \text{I.5} + \text{I.6: } BC = CG \\
 \Downarrow \\
 \text{I.34: recognize the two smaller squares} \\
 \Downarrow \\
 \text{I.43: complements } AG = GE \\
 \Downarrow \\
 \text{Sq}(AB) = \text{Sq}(AC) + \text{Sq}(CB) + 2\text{Rect}(AC, CB) \\
 \Downarrow \\
 \text{II.4.}
 \end{array}$$

This classical DAG is important because it is *not* the DAG of the Lean proof. In particular, the formal reconstruction does not locally call I.43 and does not prove the two cross rectangles equal by diagonal complements. That equality has already been absorbed into the representative-independent rectangle calculus developed after Book I.

4.4 The Geometric Identity in the Current Formalization

The public Lean theorem starts with a strict cut

$$A - C - B.$$

It receives a square on AB , two square representatives on the two parts, and two arbitrary representatives of the cross rectangle contained by AC and CB .

At the level of exact scissors terms the final conclusion has the form

$$\text{Sq}(AB) \simeq_{\text{sc}} (\text{Rect}(AC, CB) + \text{Sq}(CA)) + (\text{Rect}(AC, CB) + \text{Sq}(CB)).$$

The use of CA rather than AC in the first square is only an endpoint orientation choice. Segments are unoriented for congruence purposes, so this is the same geometric side magnitude.

The two occurrences of the cross rectangle are deliberately represented by independent quadrilaterals. The theorem therefore says more than “the same literal rectangle occurs twice in one drawing.” It says that any two representatives satisfying

```
IsRectangleContainedBy Geo ... A C C B
```

may occupy the two cross-term positions, because the rectangle layer has already proved representative independence at the exact scissors level.

A modern mnemonic is

$$x = y + z \quad \implies \quad x^2 = y^2 + z^2 + 2yz.$$

This is useful for recognizing the combinatorial pattern only. No object corresponding to yz is constructed by `HilbertSegmentMul` in the II.4 proof.

4.5 Reusable Book II Infrastructure

4.5.1 Proposition II.2: split the square on the whole

The first reusable theorem gives

$$\text{Rect}(AB, AC) + \text{Rect}(AB, CB) \simeq_{\text{sc}} \text{Sq}(AB).$$

In the current source it is called exactly as

```
have hBCA : Geo.Between B C A :=
  (HilbertOrder.between_incidence A C B hACB).2.2.2.2

-----
-- II.2:
--
--   Rect(AB,AC) + Rect(AB,CB) = Square(AB) .
-----

have hII2 :
  HilbertScissorsEq Geo
```

```

(hilbertParallelogramTerm Geo U0 U1 U2 U3 +
 hilbertParallelogramTerm Geo V0 V1 V2 V3)
(hilbertParallelogramTerm Geo A B E0 D0) :=
euclid_proposition_2_2
Geo
A B D0 E0 L0 C X0
hSquareAB
hACB
hD0L0E0
hD0X0B
hCX0L0
hII2LeftPar
hII2RightPar
U0 U1 U2 U3
V0 V1 V2 V3
hAB_AC
hAB_CB

```

The final II.4 proof uses the symmetry of this relation, so the square becomes the starting point and the two whole-by-part rectangles become the two branches to be expanded.

4.5.2 Proposition II.3: expand a whole-by-part rectangle

Proposition II.3 has the generic geometric pattern

$$\text{Rect}(XY, ZY) \simeq_{\text{sc}} \text{Rect}(XZ, ZY) + \text{Sq}(ZY)$$

for a strict division point Z on XY , with the configured rectangle cut supplied explicitly.

II.4 calls this theorem twice. The right branch uses the original orientation $A - C - B$ directly:

$$\text{Rect}(AB, CB) \simeq_{\text{sc}} \text{Rect}(AC, CB) + \text{Sq}(CB).$$

The left branch reads the same line in the reverse direction $B - C - A$:

$$\text{Rect}(BA, CA) \simeq_{\text{sc}} \text{Rect}(BC, CA) + \text{Sq}(CA).$$

After endpoint reversal and rectangle rotation, its cross term is another representative of $\text{Rect}(AC, CB)$.

4.5.3 Rectangle rotation and endpoint reversal

The only side-order normalization that is mathematically visible in II.4 is

```
rectangle_contained_by_swap
```

applied to the second cross-rectangle representative. It cyclically rotates the concrete rectangle and exchanges the roles of its containing sides while returning an exact scissors certificate for the change of presentation.

Endpoint reversal is handled by the already established congruence theorem

```
CongruentSwapSecond
```

so that AB may be read as BA , AC as CA , and CB as BC when the reversed II.3 invocation requires that orientation. These steps are representation transport, not new metric statements.

4.5.4 Exact scissors addition

Once the two II.3 branches are available, the constructor

```
HilbertScissorsEq.add
```

combines them componentwise. Transitivity then attaches the resulting four-term expression to the square through the symmetric II.2 equality.

No subtraction, cancellation, positivity, content order, or numerical area operation occurs.

4.6 Proof Architecture of the Reconstruction

4.6.1 Step 1: reverse the cut orientation for the left branch

From $A - C - B$ the proof derives $B - C - A$:

```
have hBCA : Geo.Between B C A :=
  (HilbertOrder.between_incidence A C B hACB).2.2.2.2
```

This is the only explicit betweenness reversal performed locally.

4.6.2 Step 2: apply II.2 to the square on the whole

The theorem `euclid_proposition_2_2` produces the exact decomposition of the square into the two rectangles $\text{Rect}(AB, AC)$ and $\text{Rect}(AB, CB)$. Nothing about the smaller squares is used yet.

4.6.3 Step 3: prepare the reversed left II.3 call

The first II.2 rectangle is formally contained by AB, AC , while the reversed II.3 call expects BA, CA . The same concrete rectangle is repackaged using endpoint reversal. The second final cross-rectangle representative is then rotated from the side order AC, CB to CB, AC and read, after endpoint reversal, as BC, CA .

The current source makes these transports explicit:

```
have hAB_AC_rev :
  IsRectangleContainedBy Geo
    U0 U1 U2 U3 B A C A :=
And.intro
  hAB_AC.1
  (And.intro
    (CongruentSwapSecond
      Geo U0 U1 A B hAB_AC.2.1)
```

```

      (CongruentSwapSecond
        Geo U1 U2 A C hAB_AC.2.2))

-----

-- Rotate the second cross rectangle:
--
--   Rect(AC,CB) -> Rect(CB,AC).
--
-- Endpoint reversal then reads the rotated representative as
-- Rect(BC,CA), exactly what the reversed II.3 needs.
-----

have hSwapResult :=
  rectangle_contained_by_swap
    Geo
    T20 T21 T22 T23
    A C C B
    hCross2

have hCross2Swap :=
  hSwapResult.1

have hRotateCross2 :=
  hSwapResult.2

have hCross2ForLeft :
  IsRectangleContainedBy Geo
    T21 T22 T23 T20 B C C A :=
And.intro
  hCross2Swap.1
  (And.intro
    (CongruentSwapSecond
      Geo T21 T22 C B hCross2Swap.2.1)
    (CongruentSwapSecond
      Geo T22 T23 A C hCross2Swap.2.2))

```

4.6.4 Step 4: expand the left rectangle by II.3

The reversed invocation gives

$$\text{Rect}(BA, CA) \simeq_{\text{sc}} \text{Rect}(BC, CA) + \text{Sq}(CA).$$

The rotated cross term is then transported back to the arbitrary final AC, CB representative. The exact source block is:

```

have hLeftRaw :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo U0 U1 U2 U3)
    (hilbertParallelogramTerm Geo T21 T22 T23 T20 +
      hilbertParallelogramTerm Geo C A F G) :=
euclid_proposition_2_3
  Geo
  B A D1 E1 L1 C X1
  hRectLeft
  hAE1_CA

```

```

hBCA
hD1L1E1
hD1X1A
hCX1L1
hLeftLeftPar
hLeftRightPar
F G
hSquareCA
U0 U1 U2 U3
T21 T22 T23 T20
hAB_AC_rev
hCross2ForLeft

-----
-- Return the left cross term to the original arbitrary
-- representative T20-T21-T22-T23.
-----

have hRotateCross2Back :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo T21 T22 T23 T20)
    (hilbertParallelogramTerm Geo T20 T21 T22 T23) :=
  HilbertScissorsEq.symm
    (Geo := Geo)
  hRotateCross2

have hLeftTarget :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo U0 U1 U2 U3)
    (hilbertParallelogramTerm Geo T20 T21 T22 T23 +
      hilbertParallelogramTerm Geo C A F G) :=
  HilbertScissorsEq.trans
    (Geo := Geo)
  hLeftRaw
  (HilbertScissorsEq.add
    (Geo := Geo)
    hRotateCross2Back
    (HilbertScissorsEq.refl
      (Geo := Geo)
      (hilbertParallelogramTerm Geo C A F G)))

```

4.6.5 Step 5: expand the right rectangle by II.3

The right branch needs no reversal of the whole segment. It directly produces

$$\text{Rect}(AB, CB) \simeq_{\text{sc}} \text{Rect}(AC, CB) + \text{Sq}(CB).$$

4.6.6 Step 6: add the branches and attach them to the square

The two II.3 relations are added. Finally the result is composed with the symmetric II.2 relation. The closing source is:

```

have hRight :
  HilbertScissorsEq Geo

```



```

      (hilbertParallelogramTerm Geo V0 V1 V2 V3)
      (hilbertParallelogramTerm Geo T10 T11 T12 T13 +
        hilbertParallelogramTerm Geo C B H K) :=
euclid_proposition_2_3
  Geo
  A B D2 E2 L2 C X2
  hRectRight
  hBE2_CB
  hACB
  hD2L2E2
  hD2X2B
  hCX2L2
  hRightLeftPar
  hRightRightPar
  H K
  hSquareCB
  V0 V1 V2 V3
  T10 T11 T12 T13
  hAB_CB
  hCross1

-----
-- Expand both II.2 summands simultaneously.
-----

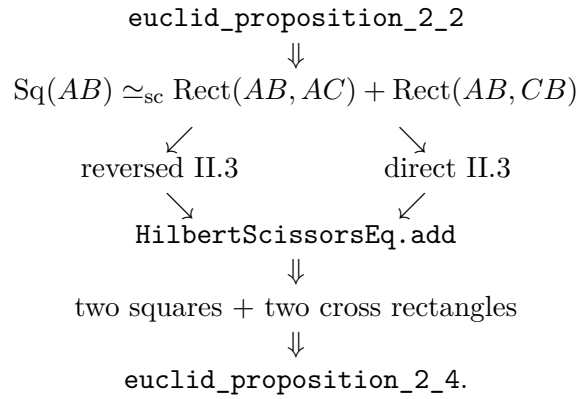
have hExpanded :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo U0 U1 U2 U3 +
      hilbertParallelogramTerm Geo V0 V1 V2 V3)
    ((hilbertParallelogramTerm Geo T20 T21 T22 T23 +
      hilbertParallelogramTerm Geo C A F G) +
      (hilbertParallelogramTerm Geo T10 T11 T12 T13 +
        hilbertParallelogramTerm Geo C B H K)) :=
  HilbertScissorsEq.add
    (Geo := Geo)
    hLeftTarget
    hRight

-----
-- Square(AB)
--   = Rect(AB,AC) + Rect(AB,CB)
--   = the four terms of II.4.
-----

exact
  HilbertScissorsEq.trans
    (Geo := Geo)
    (HilbertScissorsEq.symm
      (Geo := Geo)
      hII2)
    hExpanded

```

The actual Lean DAG is therefore



This is substantially shorter conceptually than Euclid’s local geometric DAG, but it is not an algebraic shortcut. The earlier propositions already package the relevant geometric dissections as reusable exact-scissors theorems.

4.7 Lean Formalization

The current public theorem signature is copied from the compiling `Proposition2_4.lean` source:

```

theorem euclid_proposition_2_4
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]

  (A B C : Geo.Point)

  -----
  -- II.2 diagram: square on AB cut at C.
  -----

  (D0 E0 L0 X0 : Geo.Point)
  (hSquareAB : IsSquare Geo A B E0 D0)

  (hACB : Geo.Between A C B)
  (hD0LOE0 : Geo.Between D0 L0 E0)
  (hD0X0B : Geo.Between D0 X0 B)
  (hCX0LO : Geo.Between C X0 L0)

  (hII2LeftPar :
    IsParallelogram Geo L0 D0 A C)
  (hII2RightPar :
    IsParallelogram Geo B E0 L0 C)

  -- Arbitrary representatives of Rect(AB,AC) and Rect(AB,CB).
  (U0 U1 U2 U3 : Geo.Point)
  (V0 V1 V2 V3 : Geo.Point)

  (hAB_AC :
    IsRectangleContainedBy Geo
      U0 U1 U2 U3 A B A C)

  (hAB_CB :
```

```

IsRectangleContainedBy Geo
  V0 V1 V2 V3 A B C B)

-----

-- II.3 diagram for the left term.
--
-- We read the divided whole segment in the reverse direction:
--
--   B-C-A.
--
-- Thus II.3 gives
--
--   Rect(BA,CA) = Rect(BC,CA) + Square(CA).
-----

(D1 E1 L1 X1 : Geo.Point)

(hRectLeft :
  IsRectangle Geo B A E1 D1)

(hAE1_CA :
  Geo.Congruent A E1 C A)

(hD1L1E1 : Geo.Between D1 L1 E1)
(hD1X1A : Geo.Between D1 X1 A)
(hCX1L1 : Geo.Between C X1 L1)

(hLeftLeftPar :
  IsParallelogram Geo L1 D1 B C)
(hLeftRightPar :
  IsParallelogram Geo A E1 L1 C)

-- Square on CA, i.e. on the same unoriented segment as AC.
(F G : Geo.Point)
(hSquareCA : IsSquare Geo C A F G)

-----

-- II.3 diagram for the right term:
--
--   Rect(AB,CB) = Rect(AC,CB) + Square(CB).
-----

(D2 E2 L2 X2 : Geo.Point)

(hRectRight :
  IsRectangle Geo A B E2 D2)

(hBE2_CB :
  Geo.Congruent B E2 C B)

(hD2L2E2 : Geo.Between D2 L2 E2)
(hD2X2B : Geo.Between D2 X2 B)
(hCX2L2 : Geo.Between C X2 L2)

(hRightLeftPar :
  IsParallelogram Geo L2 D2 A C)
(hRightRightPar :
  IsParallelogram Geo B E2 L2 C)

```

```

-- Square on CB.
(H K : Geo.Point)
(hSquareCB : IsSquare Geo C B H K)

-----

-- Two arbitrary representatives of the repeated cross rectangle
-- Rect(AC,CB).
-----

(T10 T11 T12 T13 : Geo.Point)
(T20 T21 T22 T23 : Geo.Point)

(hCross1 :
  IsRectangleContainedBy Geo
    T10 T11 T12 T13 A C C B)

(hCross2 :
  IsRectangleContainedBy Geo
    T20 T21 T22 T23 A C C B) :

HilbertScissorsEq Geo
  (hilbertParallelogramTerm Geo A B E0 D0)
  ((hilbertParallelogramTerm Geo T20 T21 T22 T23 +
    hilbertParallelogramTerm Geo C A F G) +
    (hilbertParallelogramTerm Geo T10 T11 T12 T13 +
    hilbertParallelogramTerm Geo C B H K)) := by

```

Several aspects of the signature deserve explicit interpretation.

First, one strict division relation $A - C - B$ is shared by the theorem, but the public statement receives *three* configured cut diagrams:

- the II.2 square-cut diagram, with points D_0, E_0, L_0, X_0 ;
- the left II.3 rectangle-cut diagram, with points D_1, E_1, L_1, X_1 ;
- the right II.3 rectangle-cut diagram, with points D_2, E_2, L_2, X_2 .

The theorem does not construct these configurations from A, C, B alone. Rather, it proves the content identity once the corresponding geometric witnesses have been supplied.

Second, $U_0U_1U_2U_3$ and $V_0V_1V_2V_3$ are arbitrary representatives of the two rectangles produced at the II.2 level. They need not be literal subrectangles of the square ABE_0D_0 .

Third, the cross terms are represented independently by $T_{10}T_{11}T_{12}T_{13}$ and $T_{20}T_{21}T_{22}T_{23}$. Both satisfy the same `IsRectangleContainedBy` predicate for AC, CB . This makes the formal “twice the rectangle” statement representative-independent rather than tied to one duplicated quadrilateral.

Fourth, the conclusion is exactly

```
HilbertScissorsEq Geo ...
```

and not the weaker

```
HilbertScissorsEquicomplementtable Geo ...
```

relation. Exactness is inherited from II.2, II.3, rectangle rotation, formal addition, and transitivity.

4.8 Relationship with Earlier Book II Propositions

4.8.1 Proposition II.1

II.4 does not import `Proposition2_1` directly. Nevertheless II.1 is the underlying distributive core because both II.2 and II.3 are themselves built from its binary rectangle split. The dependency is therefore transitive:

$$\text{II.1} \longrightarrow \text{II.2} \text{ and } \text{II.3} \longrightarrow \text{II.4}.$$

This is the first point where the investment in II.1 becomes visibly compositional at the proposition level. II.2 and II.3 were sibling specializations of the binary II.1 API; II.4 uses both siblings together.

4.8.2 Proposition II.2

II.2 supplies exactly the outer decomposition needed by II.4:

$$\text{Sq}(AB) \simeq_{\text{sc}} \text{Rect}(AB, AC) + \text{Rect}(AB, CB).$$

The current II.4 source imports II.2 explicitly and calls its public theorem once.

4.8.3 Proposition II.3

II.3 supplies the expansion of each whole-by-part rectangle. The current proof calls the same public theorem twice, once after reversing the line orientation and once directly. Thus II.4 is a genuine composition theorem over the public Book II API rather than a restatement of one earlier helper.

4.9 Relationship with Book I Infrastructure

Euclid's own local proof depends explicitly on I.46, I.31, I.29, I.5, I.6, I.34, and I.43. The formal II.4 theorem does not replay any of those arguments locally. Their geometric content has entered the current project through the established square, parallel, parallelogram, rectangle, and scissors layers and through the already proved II.2 and II.3 theorems.

The contrast with I.43 is especially instructive. Euclid needs I.43 to prove that the two complements AG and GE in one concrete square are equal. Lean instead receives two arbitrary cross-rectangle representatives and normalizes one of them with `rectangle_contained_by_swap`. Representative independence is already built into the rectangle API, so no local complement theorem is required.

The formal proof is therefore not a transcription of Euclid's diagram. It is a proof over a stronger reusable library whose theorems were themselves proved synthetically from the established Hilbert/Book I geometry.

4.10 Formalization Notes and Hidden Geometric Data

The polished identity hides a substantial amount of explicit positional data. The current signature contains:

- $A - C - B$, fixing the strict order of the original segment;
- a complete square-cut configuration for the II.2 call;
- two complete rectangle-cut configurations for the two II.3 calls;
- three diagonal/cut intersection witnesses X_0, X_1, X_2 ;
- six parallelogram hypotheses certifying the two pieces in each cut;
- one square on the whole and one square on each part;
- two arbitrary II.2 rectangle representatives;
- two arbitrary representatives of the repeated cross rectangle.

The proof then adds two representational obligations not visible in Euclid's statement:

- reverse $A - C - B$ to $B - C - A$ for the left II.3 call;
- rotate one cross-rectangle representative and reverse segment endpoints so that the ordered side data match the reversed II.3 interface.

There is also useful negative information. No `SameRay`, `SameSide`, or `OppositeSide` datum occurs in the public II.4 proof. No new right-angle propagation theorem, no new parallel theorem, and no new scissors constructor is introduced. Those obligations are encapsulated inside the imported rectangle and Book II layers.

4.11 Geometry, Content, Representation, and Later Arithmetic

4.11.1 Geometry

The geometry consists of one segment cut $A - C - B$, one square on the whole, two squares on the parts, and three configured rectangle dissections. The classical source packages these ideas in one square diagram; the Lean theorem packages them as three reusable instances of already formalized cut theorems.

4.11.2 Content / scissors

The conclusion is exact `HilbertScissorsEq`. The proof only uses reflexivity, symmetry, transitivity, compatibility with addition, and exact scissors certificates supplied by II.2, II.3, and rectangle rotation. General equicomplementability is not required.

4.11.3 Representation

Representation is the genuinely nontrivial local layer. Endpoint order on the containing segments must be normalized for the reversed II.3 branch, and one cross rectangle must be cyclically rotated before it can be used in that branch. The final theorem nevertheless permits two completely independent representatives of $\text{Rect}(AC, CB)$.

This is the precise formal content of “twice the rectangle” in the current architecture: two scissors terms representing the same geometric rectangle magnitude occur as separate summands.

4.11.4 Later arithmetic / numerical area

Neither later layer is used. The proof imports no segment multiplication module, chooses no unit segment, proves no distributive law for a product of segment classes, and evaluates no real-valued area function.

The formula

$$(y + z)^2 = y^2 + z^2 + 2yz$$

should therefore remain explicitly labelled as the modern algebraic shadow of the synthetic construction.

4.12 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and therefore belongs to the same Euclidean project layer as II.1–II.3.

The current `Proposition2_4.lean` file introduces no proposition-local `axiom`, `sorry`, or `admit`. The theorem was compiled successfully in the project. The supplied transitive axiom audit is:

```
#print axioms Geometry.euclid_proposition_2_4

'Geometry.euclid_proposition_2_4' depends on axioms:
[propext, Classical.choice, bookZero_nullSegment1, Quot.sound]
```

Accordingly II.4 should not be described as “axiom-free.” The precise statement is that it introduces no new proposition-specific axiom and no `sorry`; its transitive dependencies are exactly the four declarations listed above. In particular the list contains no `sorryAx`.

The compile result reported for the current theorem is clean. A separate full `lake build` result was not part of the documentation input used here, so the status table below does not claim that stronger check.

4.13 Hartshorne’s Semantic Control

Hartshorne’s Section 22 supplies the appropriate general semantic control for Book II: Euclid’s equalities of plane figures can be interpreted through equal content without first assigning numerical areas [5]. He also explains why the algebraic-looking manipulations of rectangles and squares may be regarded as a kind of geometrical algebra.

II.4 is a particularly vivid example of that interpretation. Its modern shadow is a quadratic identity, but Euclid’s objects are still a large square, two smaller squares, and two rectangles. The current formalization sharpens the configured result from general equal content to exact scissors equality, because every step is built from exact dissection and exact representation transport.

Hartshorne’s “Brief Euclid” explicitly lists II.4 in the form that the square on the whole line equals the squares on its two segments plus twice the rectangle on the two segments [5]. Thus II.4 has a more direct Hartshorne-side textual counterpart than II.2 or II.3. The surrounding Section 22 discussion still supplies the essential semantic interpretation: Euclid’s equality is read as equal content, not as prior equality of numerical areas.

The stronger exact-scissors certificate remains a fact about the present Lean architecture. It should not be attributed to Hartshorne’s general content relation.

4.14 Hilbert’s Layering

Hilbert provides the complementary architectural warning. His plane-content theory and his later arithmetic of segments are distinct layers. Segment multiplication is introduced only later, using a fixed unit segment and parallel constructions, after which commutativity and distributivity become actual arithmetic theorems [2].

II.4 may look like the first place where one should invoke those later laws, because the shadow

$$(y + z)^2 = y^2 + z^2 + 2yz$$

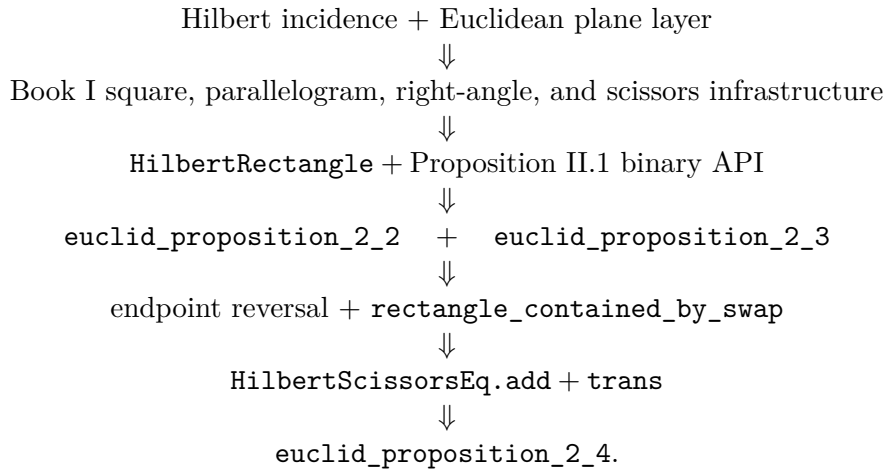
contains both distributivity and commutativity. The compiling source shows that this move is unnecessary. The proof stays below segment arithmetic:

rectangle/square geometry
 $\Downarrow$
 exact II.2 and II.3 scissors identities
 $\Downarrow$
 geometric representation transport
 $\Downarrow$
 exact II.4 scissors identity.

This is a useful architectural checkpoint. Even a full quadratic-looking Book II identity can be derived from the geometric rectangle calculus without crossing into Hilbert’s later segment field.

4.15 Dependency Summary

The actual Lean dependency chain is



The classical local chain is different:

$$\text{I.46} + \text{I.31} + \text{I.29} + \text{I.5} + \text{I.6} + \text{I.34} + \text{I.43} \longrightarrow \text{II.4.}$$

New geometric axiom in II.4:	no
New content axiom in II.4:	no
New proposition-local helper theorem:	no
New reusable helper theorem:	no
New <code>HilbertRectangle</code> theorem introduced by II.4:	no
New Interface / Book Zero / Scissors theorem:	no
Reuses earlier Book II propositions:	yes, II.2 and II.3
Direct import of II.1:	no; inherited through II.2 and II.3
Direct side-order transport used:	yes, <code>rectangle_contained_by_swap</code>
Endpoint-reversal transport used:	yes
Exact <code>HilbertScissorsEq</code> :	yes
General equicomplementability required:	no
Numerical area:	no
Segment multiplication:	no
Public Lean theorem compile:	yes, clean
Full relevant <code>lake build</code> :	not separately reported here
Proposition-local <code>sorry/admit</code> :	no
<code>sorryAx</code> in reported axiom list:	no

4.16 Conclusion

Proposition II.4 is the first strong compositional test of the Book II API. Euclid proves the theorem inside one carefully constructed square: two subfigures are shown to be squares, and I.43 identifies the two remaining complements. The current Lean proof instead works one level higher. It takes II.2 as the outer square decomposition and applies II.3 to each of the two resulting rectangles.

The only local difficulty is representational. One II.3 call uses the reversed order $B - C - A$, and one cross rectangle must be rotated so that its ordered containing sides match that call. Once these transports are made, `HilbertScissorsEq.add` and transitivity close the theorem.

Consequently II.4 requires no new rectangle theorem, no new scissors theorem, no new proposition-local helper, and no segment multiplication. Its apparent quadratic algebra is already present in the synthetic rectangle calculus as a composition of exact geometric decompositions. The reported axiom audit adds no new proposition-specific debt and contains no `sorryAx`.

Chapter 5

Proposition II.5

5.1 Introduction

Proposition II.5 is the first Book II theorem in the present reconstruction to combine a *midpoint condition* with the rectangle/scissors calculus. Let C bisect AB , and let D be a second cut point. In the ordering chosen by the Lean theorem,

$$A - D - C - B,$$

so that C is the midpoint of AB and D lies in the left half. The theorem asserts, in geometric content language,

$$\text{Rect}(AD, DB) + \text{Sq}(CD) = \text{Sq}(CB).$$

The familiar modern algebraic shadow is obtained by writing the half-length as a and the displacement of the unequal cut from the midpoint as x :

$$(a - x)(a + x) + x^2 = a^2.$$

This formula is only an interpretive mnemonic. The Lean proof does not define multiplication of segment lengths and does not evaluate numerical area. Its objects are concrete rectangle and square representatives, while equality is expressed by exact scissors equivalence.

The current source begins with

```
import CGJteamLab.Proposition2_4
```

but this import fact should not be confused with the proof DAG. The proof does not invoke `euclid_proposition_2_4`. It directly reuses the binary rectangle theorem from II.1 and Proposition II.3, together with the reusable rectangle representation API. Thus II.5 provides an especially useful example of the difference between module dependency and mathematical proof dependency.

5.2 Euclid's Statement and Classical Configuration

5.2.1 The statement

In the notation used here, Euclid's proposition may be stated as follows. A straight line AB is divided equally at C and unequally at D . Then the rectangle contained by AD and DB , together

with the square on CD , is equal in content to the square on the half CB [1].

The phrase “rectangle contained by AD, DB ” names a plane figure whose adjacent sides have those magnitudes. It is not a product already formed in a segment field. Likewise, the addition in the displayed identity describes the content of two figures taken together.

5.2.2 The classical diagram

The principal figure below follows the standard Euclidean configuration with the unequal cut D chosen in the right half, so its baseline order is $A - C - D - B$. This is the mirror version of the Lean convention $A - D - C - B$; the content identity is unchanged. Euclid erects the square $CEFB$ on the half CB , joins BE , draws the parallel through D , then the horizontal through its intersection H with the diagonal, and finally the parallel through A [1].

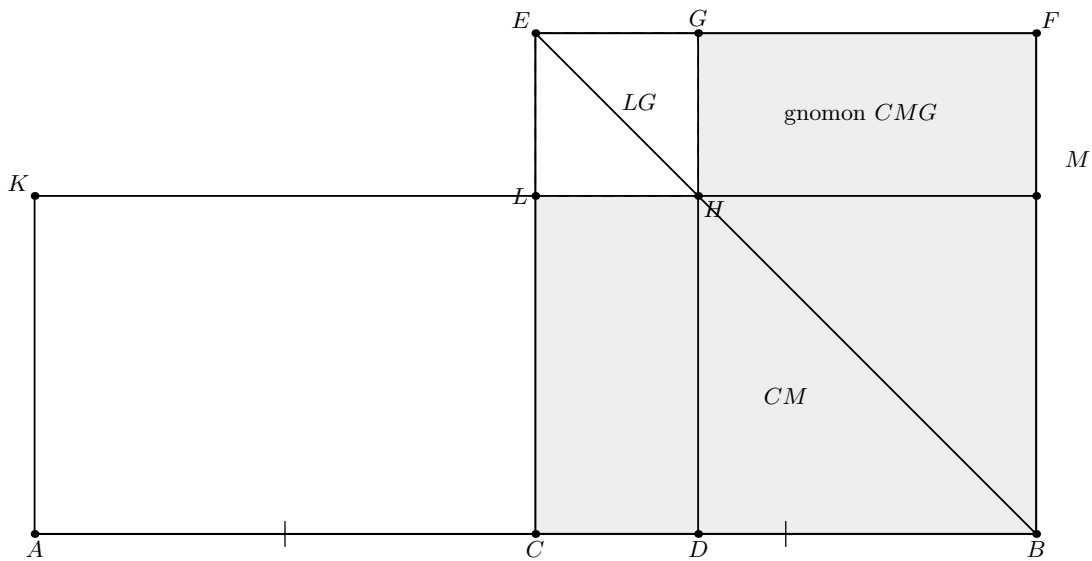


Figure 5.1: The classical mechanism of Euclid II.5. The square $CEFB$ is built on the half CB . The point D is the unequal cut, H lies on diagonal BE , and LG is the small square on CD . The remaining part of $CEFB$ is the gnomon used in Euclid’s content argument. The formal Lean proof does not introduce a primitive gnomon object.

5.3 Classical Proof

The classical proof uses the gnomon structure of Figure 5.1. The diagonal BE and the parallels create complementary parallelograms. By I.43 the complements CH and HF are equal. Adding the same rectangle DM to the relevant figures yields an equality in which the rectangle CM can be replaced by AL : these are equal parallelograms on equal bases AC and CB and in the same parallels, using I.36 and the midpoint condition $AC = CB$ [1].

The construction also gives $DH = DB$ by the square-of-a-sum geometry already available at II.4 (in Euclid’s classical dependency chain). Consequently the figure AH is the rectangle contained by the unequal parts AD and DB . Euclid identifies that rectangle with the gnomon obtained by removing the small corner square LG from the square $CEFB$. Finally LG is the square on the segment CD , and adjoining it to the gnomon recovers the entire square on CB .

Thus the classical local DAG is schematically

$$\begin{array}{c}
 \text{I.43: equal complements} \quad + \quad \text{I.36: equal parallelograms on equal bases} \\
 \text{together with the II.4 corollary} \\
 \Downarrow \\
 \text{rectangle on } AD, DB = \text{gnomon in the square on } CB \\
 \Downarrow \\
 \text{add the square on } CD \\
 \Downarrow \\
 \text{Rect}(AD, DB) + \text{Sq}(CD) = \text{Sq}(CB).
 \end{array}$$

This is *not* the Lean DAG. In particular, the current formal proof does not reconstruct the classical gnomon, does not call I.43 locally, and does not call II.4 despite importing the II.4 module.

5.4 The Geometric Identity in the Current Formalization

The public Lean theorem chooses the strict order

$$A - D - C - B$$

through two hypotheses:

```
(hMidC : HilbertIsMidpoint Geo C A B)
(hADC : Geo.Between A D C)
```

The midpoint hypothesis supplies $A - C - B$ and $AC \cong CB$. From $A - D - C$ and $A - C - B$, a private order helper derives $D - C - B$ and $A - D - B$. Thus the full four-point order is a theorem of the hypotheses, not an additional assumption.

At the exact-scissors level the target is

$$\text{Rect}(DB, AD) + \text{Sq}(DC) \simeq_{\text{sc}} \text{Sq}(CB).$$

The reversed order DB, AD rather than AD, DB is immaterial at the geometric rectangle level because side-order exchange is itself proved by rectangle rotation/transport. It is not treated as primitive commutativity of a segment product.

A notable semantic point is that no gnomon type occurs in the source. The classical gnomon is represented extensionally by formal sums of rectangle and square scissors terms. The proof then rearranges and composes those sums by `HilbertScissorsEq.add`, symmetry, transitivity, and the existing rectangle transport lemmas.

5.5 Reusable Book II Infrastructure

5.5.1 The binary rectangle split from II.1

The theorem `euclid_proposition_2_1_two` is used twice. First it splits DB at C , with AD held as the other containing side:

$$\text{Rect}(DB, AD) \simeq_{\text{sc}} \text{Rect}(DC, AD) + \text{Rect}(CB, AD).$$

Later the same binary theorem, now applied to AC split at D , assembles

$$\text{Rect}(CB, AD) + \text{Rect}(CB, DC) \simeq_{\text{sc}} \text{Rect}(CB, AC),$$

after geometric side-order transports.

5.5.2 Proposition II.3

Proposition II.3 supplies the first quadratic-looking decomposition:

$$\text{Rect}(AC, DC) \simeq_{\text{sc}} \text{Rect}(AD, DC) + \text{Sq}(DC).$$

The current source calls `euclid_proposition_2_3` directly. The midpoint congruence $AC \cong CB$ is then used to transport the left-hand rectangle to a representative contained by CB, DC .

5.5.3 Rectangle representation and transport

The proof relies on the established Book II rectangle API rather than numerical products. In particular it uses

- `rectangle_transport_scissorsEq` to change containing segments along segment congruences;
- `rectangle_contained_by_comm` and `rectangle_contained_by_swap` to exchange the two adjacent side roles geometrically;
- `rectangle_contained_by_unique` to compare arbitrary representatives of the same contained-by data;
- `square_is_rectangle_contained_by_side` to view a square as a rectangle contained by one side twice.

5.6 Proof Architecture of the Reconstruction

The production file preserves the private helper names created during the incremental proof search. The names contain `testXX`, but all these lemmas are `private`; they are implementation history, not public API. Their mathematical roles are stable and form the following chain.

5.6.1 Step 1: recover the full order and midpoint congruence

`proposition2_5_test01_order` derives

$$D - C - B, \quad A - D - B, \quad AC \cong CB, \quad CA \cong CB.$$

This is the only place where the four-point order is consolidated.

5.6.2 Step 2: apply II.3 on the left half

`proposition2_5_test02_left_decomposition` proves

$$\text{Rect}(AC, DC) \simeq_{\text{sc}} \text{Rect}(AD, DC) + \text{Sq}(DC).$$

5.6.3 Step 3: transport the midpoint side

`proposition2_5_test03_midpoint_rectangle_transport` uses $AC \cong CB$ to prove

$$\text{Rect}(AC, DC) \simeq_{\text{sc}} \text{Rect}(CB, DC).$$

By symmetry and transitivity, `proposition2_5_test04_first_assembled_identity` obtains

$$\text{Rect}(CB, DC) \simeq_{\text{sc}} \text{Rect}(AD, DC) + \text{Sq}(DC).$$

5.6.4 Step 4: split the unequal part DB

`proposition2_5_test05_db_decomposition` applies the binary II.1 theorem to $D - C - B$:

$$\text{Rect}(DB, AD) \simeq_{\text{sc}} \text{Rect}(DC, AD) + \text{Rect}(CB, AD).$$

The next two helpers use geometric commutativity to normalize the first term:

$$\text{Rect}(DB, AD) \simeq_{\text{sc}} \text{Rect}(AD, DC) + \text{Rect}(CB, AD).$$

5.6.5 Step 5: the scissors bridge

The private lemma `proposition2_5_test08_scissors_bridge` is an abstract content calculation. Given

$$P \simeq X + Y, \quad Z \simeq X + S,$$

it proves

$$P + S \simeq Y + Z.$$

Crucially, this is *not* a cancellation principle. It is obtained by adding exact scissors equalities, reassociating and commuting the formal multiset sum, and using transitivity. Applying it to the two preceding identities gives

$$\boxed{\text{Rect}(DB, AD) + \text{Sq}(DC) \simeq_{\text{sc}} \text{Rect}(CB, AD) + \text{Rect}(CB, DC)}.$$

5.6.6 Step 6: assemble the right side

`proposition2_5_test09_assemble_right_side` rotates the two right-hand rectangle representatives, applies the binary II.1 split to $A - D - C$, and rotates the result back. It proves

$$\text{Rect}(CB, AD) + \text{Rect}(CB, DC) \simeq_{\text{sc}} \text{Rect}(CB, AC).$$

5.6.7 Step 7: recognize the half-square

Finally `proposition2_5_test10_half_rectangle_is_square` uses $AC \cong CB$ and representative uniqueness to identify

$$\text{Rect}(CB, AC) \simeq_{\text{sc}} \text{Sq}(CB).$$

This is where the midpoint hypothesis enters a second time: first it controlled order, and now it turns the final rectangle into the square on the half.

5.7 Lean Formalization

5.7.1 The public theorem signature

The following is the current compiling public signature. It is intentionally large because the theorem receives the concrete rectangle-cut configurations and representatives required by the earlier Book II APIs rather than hiding them behind choice constructions.

```

theorem euclid_proposition_2_5
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]

  (A B C D : Geo.Point)
  (hMidC : HilbertIsMidpoint Geo C A B)
  (hADC : Geo.Between A D C)

  -----

  -- Data used by test 04 inside test 08.
  -----

  (E4 H4 L4 X4 : Geo.Point)
  (hRect4 : IsRectangle Geo A C H4 E4)
  (hCH_DC : Geo.Congruent C H4 D C)
  (hELE4 : Geo.Between E4 L4 H4)
  (hEXC4 : Geo.Between E4 X4 C)
  (hDXL4 : Geo.Between D X4 L4)
  (hLeftPar4 : IsParallelogram Geo L4 E4 A D)
  (hRightPar4 : IsParallelogram Geo C H4 L4 D)

  -- Square on DC.
  (F G : Geo.Point)
  (hSquareDC : IsSquare Geo D C F G)

  -- Rect(AC,DC).
  (R0 R1 R2 R3 : Geo.Point)
  (hAC_DC :
    IsRectangleContainedBy Geo
      R0 R1 R2 R3 A C D C)

  -- Common Rect(AD,DC).
  (T0 T1 T2 T3 : Geo.Point)
  (hAD_DC :
    IsRectangleContainedBy Geo
      T0 T1 T2 T3 A D D C)

  -- Rect(CB,DC).
  (Z0 Z1 Z2 Z3 : Geo.Point)
  (hCB_DC :
    IsRectangleContainedBy Geo
      Z0 Z1 Z2 Z3 C B D C)

  -----

  -- Data used by test 07 inside test 08.
  -----

  (E7 H7 L7 X7 : Geo.Point)

```



```

(hRect7 : IsRectangle Geo D B H7 E7)
(hBH_AD : Geo.Congruent B H7 A D)
(hELE7 : Geo.Between E7 L7 H7)
(hEXB7 : Geo.Between E7 X7 B)
(hCXL7 : Geo.Between C X7 L7)
(hLeftPar7 : IsParallelogram Geo L7 E7 D C)
(hRightPar7 : IsParallelogram Geo B H7 L7 C)

-- Rect(DB,AD), the unequal-parts rectangle.
(P0 P1 P2 P3 : Geo.Point)
(hDB_AD :
  IsRectangleContainedBy Geo
    P0 P1 P2 P3 D B A D)

-- Rect(DC,AD), before commutativity normalization.
(Q0 Q1 Q2 Q3 : Geo.Point)
(hDC_AD :
  IsRectangleContainedBy Geo
    Q0 Q1 Q2 Q3 D C A D)

-- Rect(CB,AD).
(Y0 Y1 Y2 Y3 : Geo.Point)
(hCB_AD :
  IsRectangleContainedBy Geo
    Y0 Y1 Y2 Y3 C B A D)

-----

-- Data for test 09:
-- assemble Rect(CB,AD) + Rect(CB,DC) into Rect(CB,AC).
-----

(E9 H9 L9 X9 : Geo.Point)
(hRect9 : IsRectangle Geo A C H9 E9)
(hCH_CB9 : Geo.Congruent C H9 C B)
(hELE9 : Geo.Between E9 L9 H9)
(hEXC9 : Geo.Between E9 X9 C)
(hDXL9 : Geo.Between D X9 L9)
(hLeftPar9 : IsParallelogram Geo L9 E9 A D)
(hRightPar9 : IsParallelogram Geo C H9 L9 D)

-- Rect(CB,AC).
(W0 W1 W2 W3 : Geo.Point)
(hCB_AC :
  IsRectangleContainedBy Geo
    W0 W1 W2 W3 C B A C)

-----

-- Data for test 10: square on CB.
-----

(S0 S1 S2 S3 : Geo.Point)
(hSquareCB : IsSquare Geo S0 S1 S2 S3)
(hS01_CB : Geo.Congruent S0 S1 C B) :

HilbertScissorsEq Geo
  (hilbertParallelogramTerm Geo P0 P1 P2 P3 +
    hilbertParallelogramTerm Geo D C F G)
  (hilbertParallelogramTerm Geo S0 S1 S2 S3) := by

```

Several layers of data are visible in the signature.

1. A, B, C, D and the midpoint/betweenness hypotheses encode the one-dimensional configuration.
2. The groups labelled in the source as data for tests 04, 07, and 09 are concrete rectangle-cut witnesses needed by II.3 or II.1.
3. F, G give a square on DC .
4. $P_0P_1P_2P_3$ represents the rectangle contained by DB, AD .
5. $W_0W_1W_2W_3$ represents the intermediate rectangle contained by CB, AC .
6. $S_0S_1S_2S_3$ is an arbitrary square representative whose side is congruent to CB .

Thus the theorem proves representative-independent content equality for supplied geometric witnesses; it does not claim that all of those witnesses are constructed internally from four points alone.

5.7.2 The final three-stage assembly

After all private helpers have been established, the public theorem is short in conceptual structure. The current source builds exactly three equations:

1. h08: unequal-parts rectangle plus the DC square equals two rectangles with common side CB ;
2. h09: those two rectangles assemble into the rectangle $\text{Rect}(CB, AC)$;
3. h10: midpoint congruence turns that rectangle into the square on CB .

The final transitivity block is:

```

have h08 :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo P0 P1 P2 P3 +
     hilbertParallelogramTerm Geo D C F G)
    (hilbertParallelogramTerm Geo Y0 Y1 Y2 Y3 +
     hilbertParallelogramTerm Geo Z0 Z1 Z2 Z3) :=
  proposition2_5_test08_core_identity
  Geo
  A B C D
  hMidC hADC
  E4 H4 L4 X4
  hRect4 hCH_DC hELE4 hEXC4 hDXL4
  hLeftPar4 hRightPar4
  F G hSquareDC
  R0 R1 R2 R3
  hAC_DC
  T0 T1 T2 T3
  hAD_DC
  Z0 Z1 Z2 Z3
  hCB_DC
  E7 H7 L7 X7

```

```

hRect7 hBH_AD hELE7 hEXB7 hCXL7
hLeftPar7 hRightPar7
P0 P1 P2 P3
hDB_AD
Q0 Q1 Q2 Q3
hDC_AD
Y0 Y1 Y2 Y3
hCB_AD

-----

-- test09:
-- Rect(CB,AD) + Rect(CB,DC) = Rect(CB,AC).
-----

have h09 :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo Y0 Y1 Y2 Y3 +
     hilbertParallelogramTerm Geo Z0 Z1 Z2 Z3)
    (hilbertParallelogramTerm Geo W0 W1 W2 W3) :=
  proposition2_5_test09_assemble_right_side
  Geo
  A B C D
  hADC
  E9 H9 L9 X9
  hRect9 hCH_CB9 hELE9 hEXC9 hDXL9
  hLeftPar9 hRightPar9
  Y0 Y1 Y2 Y3
  hCB_AD
  Z0 Z1 Z2 Z3
  hCB_DC
  W0 W1 W2 W3
  hCB_AC

-----

-- test10:
-- Rect(CB,AC) = Square(CB).
-----

have h10 :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo W0 W1 W2 W3)
    (hilbertParallelogramTerm Geo S0 S1 S2 S3) :=
  proposition2_5_test10_half_rectangle_is_square
  Geo
  A B C D
  hMidC hADC
  W0 W1 W2 W3
  hCB_AC
  S0 S1 S2 S3
  hSquareCB
  hS01_CB

-----

-- Full II.5.
-----

exact
  HilbertScissorsEq.trans

```

```

(Geo := Geo)
h08
(HilbertScissorsEq.trans
  (Geo := Geo)
  h09

```

The formal result is therefore exact `HilbertScissorsEq`, not the weaker relation `HilbertScissorsEquicomplementable`, and not equality of real numbers.

5.8 Relationship with Earlier Book II Propositions

5.8.1 Proposition II.1

II.1 is the basic additive engine of the proof. Its binary theorem is called directly in the decomposition of DB and again in the assembly of AC . Thus II.5 genuinely reuses the rectangle split calculus rather than reproving the corresponding dissections from Book I geometry.

5.8.2 Proposition II.3

II.3 is also called directly. It supplies the identity

$$\text{Rect}(AC, DC) = \text{Rect}(AD, DC) + \text{Sq}(DC),$$

at the exact scissors level. This is the step that inserts the square on the difference between the two section points.

5.8.3 Proposition II.4

The source file imports `CGJteamLab.Proposition2_4`, so II.4 is a direct *module* dependency. Nevertheless the theorem `euclid_proposition_2_4` is not invoked in the proof term. This sharply contrasts with Euclid’s classical proof, where a corollary of II.4 participates in the identification of the relevant rectangle and small square.

This distinction should be recorded explicitly:

module chain: II.4 module $\rightarrow$ II.5 module,
 Lean theorem calls: II.1 binary API + II.3 + rectangle API.

5.9 Relationship with Book I Infrastructure

The Book I geometry is present below the reusable rectangle layer. Rectangles, squares, right angles, parallel sides, parallelograms, and finite scissors terms all ultimately depend on the established Hilbert/Book I development. II.5 does not locally replay Euclid’s diagonal-complement argument from I.43. Instead, the effect of those earlier geometric facts has already been packaged into `HilbertRectangle`, the scissors API, and the earlier Book II theorems.

This is an architectural gain: a proposition whose classical proof is diagram specific becomes a composition of stable content-level interfaces.

5.10 Formalization Notes and Hidden Geometric Data

5.10.1 Order orientation

The classical diagram commonly places the unequal cut in the right half $A - C - D - B$, whereas the Lean theorem fixes $A - D - C - B$. These are mirror cases. The source makes the chosen orientation explicit through `hADC` and then derives $D - C - B$ from the midpoint betweenness. No appeal to a picture is allowed to supply this ordering silently.

5.10.2 Midpoint as geometry, not arithmetic

`HilbertIsMidpoint` contributes both betweenness and segment congruence. The proof uses those components separately. Betweenness creates the required subsegment orders; congruence transports rectangle sides and eventually recognizes the square on the half. There is no variable standing for a numerical half-length.

5.10.3 Representative independence

The objects named “the rectangle contained by” particular segments are concrete quadrilateral representatives. Different helpers may receive different representatives of the same pair of side magnitudes. `rectangle_transport_scissorsEq`, `rectangle_contained_by_unique`, and the side-order transport theorems are therefore mathematically substantive parts of the formal statement, not mere syntactic rewrites.

5.10.4 No gnomon primitive

Euclid’s proof naturally names a gnomon. The present library does not define a `GnomonTerm` or another primitive gnomon object for II.5. No such new abstraction is needed here: the required content is expressed as a formal sum of scissors terms. The private scissors bridge then performs the exact content rearrangement that the classical gnomon packages visually.

5.10.5 No cancellation axiom

At an intermediate stage the two identities contain the common rectangle $\text{Rect}(AD, DC)$. It might be tempting to introduce a cancellation principle for content. The actual source deliberately avoids this. The scissors bridge combines equalities additively and rearranges formal sums, so no new positivity or cancellation theorem is required.

5.11 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and the production file introduces no new proposition-specific `axiom`, `sorry`, or `admit`. The user-verified module build is clean. The reported transitive axiom audit is

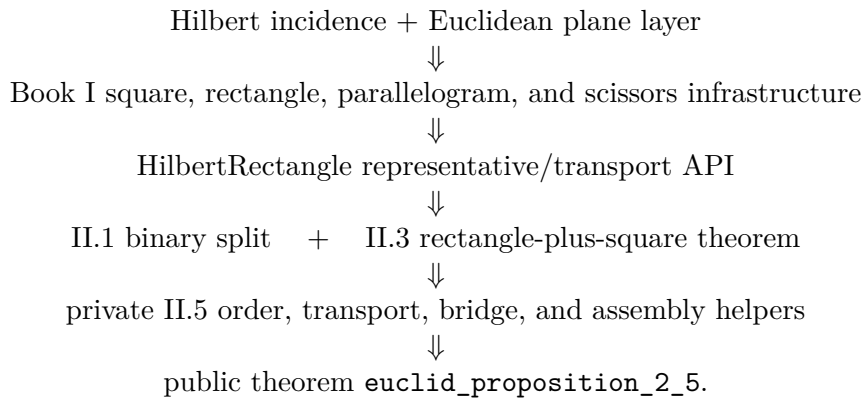
```
#print axioms Geometry.euclid_proposition_2_5

'Geometry.euclid_proposition_2_5' depends on axioms: [propext,
Classical.choice,
Geometry.bookZero_nullSegment1,
Quot.sound]
```

Accordingly II.5 should not be described as “axiom-free.” The precise claim is that II.5 introduces no new proposition-specific axiom and no `sorry`; its transitive dependencies are the four declarations shown above, and the list contains no `sorryAx`.

5.12 Dependency Summary

The actual Lean dependency chain is best displayed as



The classical local chain is different:

$$\text{I.43} + \text{I.36} + \text{II.4 corollary} \longrightarrow \text{gnomon identification} \longrightarrow \text{II.5.}$$

The formal proof is therefore not a line-by-line translation of the classical one. It reconstructs the same geometric content through a different, reusable DAG.

New geometric axiom in II.5:	no
New content axiom in II.5:	no
New proposition-local helper theorem:	yes, private helpers
New reusable public helper theorem:	no
New <code>HilbertRectangle</code> theorem:	no
New Interface / Book Zero / Scissors theorem:	no
Directly reuses earlier Book II theorem:	yes, II.1 and II.3
Imports II.4 module:	yes
Calls <code>euclid_proposition_2_4</code> :	no
Uses exact <code>HilbertScissorsEq</code> :	yes
Uses general equicomplementability:	no
Uses numerical area:	no
Uses segment multiplication:	no
Public theorem compiles:	yes
Module build:	clean (user verified)
Proposition-local <code>sorry/admit</code> :	no
Transitive <code>sorryAx</code> :	absent in reported audit

5.13 Conclusion

Proposition II.5 is an important architectural checkpoint for Book II. The classical proof organizes the equality around a gnomon inside the square on the half. The formal proof shows that a separate gnomon primitive is not needed to recover the theorem: the same content can be assembled from the binary II.1 rectangle split, II.3, midpoint congruence, exact rectangle transport, and the additive structure of `HilbertScissorsEq`.

The result also clarifies the role of the modern quadratic identity. It is a useful shadow for recognizing the proposition, but the verified argument stays synthetic throughout. Geometry supplies the order and congruences; representation supplies rectangles and squares; scissors equality supplies the content calculus. Numerical area and segment multiplication remain outside the proof.

Appendix A

Greenberg: Foundational and Semantic Control

.1 Euclid I.1–I.20 in Greenberg’s Hilbert-Style Development

.1.1 Purpose of the comparison

This appendix records the status of Euclid’s first twenty propositions in Marvin J. Greenberg’s Hilbert-style development of elementary geometry. The comparison is deliberately stricter than a list of mathematically related results. For each proposition I.1–I.20, the question is whether Greenberg provides a corresponding result and, if so, in what form: a complete synthetic proof, an axiom, a proposition whose proof is left as an exercise, a proof outline or hint, or a proof that already uses numerical measurement.

This distinction is important for the present formalization. The fact that the mathematical content of a Euclidean proposition occurs in a modern axiomatic treatment does not imply that a complete synthetic proof, suitable for direct reconstruction in Lean, is actually given there.

.1.2 Comparison table

Euclid	Greenberg counterpart	Status in Greenberg	Relation to the present formalization
I.1	Equilateral triangle construction; explicitly discussed as Euclid’s Proposition 1.	Euclid’s construction is discussed together with its continuity gap. Greenberg repairs the existence argument by means of circular continuity.	The proposition is not simply a consequence of incidence, order, and congruence alone in Greenberg’s presentation. It therefore has a different foundational status from many later propositions.
I.2	Segment laying-off; Congruence Axiom C-1.	The essential construction is incorporated into the congruence axioms rather than repoved as Euclid’s proposition.	In the Hilbert layer this operation belongs naturally to the primitive segment-construction infrastructure.
I.3	Cutting off a segment equal to a given shorter segment.	Its content is distributed between segment construction and segment ordering; no separate reconstruction of Euclid I.3 is required in Greenberg’s organization.	The formal proof is assembled from lower-level construction and order lemmas rather than imported as a named Euclidean theorem.
I.4	SAS triangle congruence.	SAS is Congruence Axiom C-6. Greenberg explicitly discusses why Euclid’s superposition argument is not accepted as an axiomatic proof.	In a Hilbert-style formalization I.4 is therefore foundational rather than a theorem derived from the preceding Euclidean propositions.
I.5	Proposition 3.10: base angles of an isosceles triangle.	A complete synthetic proof is given from SAS.	This is a direct and useful Hilbert-style counterpart of Euclid I.5.
I.6	Proposition 3.18: converse of the isosceles base-angle theorem.	The proposition is stated, but its proof is left as an exercise.	Greenberg confirms the theorem and its place in the theory, but does not supply a complete proof for direct transcription into Lean.
I.7	Uniqueness phenomenon for triangles constructed on a fixed base.	No direct proposition-by-proposition reconstruction of Euclid I.7 was identified. Related uniqueness information is built into the congruence and triangle-laying-off machinery.	The formalization must expose the exact uniqueness statement required by the Euclidean proof rather than relying on a named Greenberg counterpart.
I.8	Proposition 3.22: SSS criterion for congruence.	SSS is stated as a proposition. Its proof is assigned as an exercise, with a substantial hint based on the corollary to SAS and a case analysis.	Greenberg supplies the mathematical route but not a complete printed synthetic proof.
I.9	Angle bisector construction; Proposition 4.4.	The result occurs in the neutral-geometry development, but the proof is assigned to the reader.	The existence of the construction is standard in the theory, while the formal reconstruction still requires the missing synthetic details.
I.10	Midpoint construction; Proposition 4.3 and associated exercise.	Greenberg supplies a detailed step-by-step proof scheme in the exercises, with the individual steps to be justified.	This is close to a proof blueprint, but it is not presented as a complete finished proof in the main text.
I.11–I.12	Proposition 3.16: existence of a perpendicular through a point.	A complete synthetic proof is given, treating separately the cases in which the point lies on the line and does not lie on the line.	The single Hilbert-style proposition covers the geometric content split by Euclid into I.11 and I.12.
I.13–I.14	Supplementary angles, rays, angle addition/subtraction, and related order results.	The required content is distributed through the congruence and betweenness theory rather than organized as two direct Euclidean counterparts.	These propositions illustrate why a proposition-number correspondence is insufficient: the modern proof infrastructure reorganizes the underlying elementary geometry.

.1.3 Result of the query

The comparison shows that Greenberg does *not* contain a continuous sequence of twenty complete synthetic proofs corresponding to Euclid I.1–I.20.

Instead, the first twenty propositions are distributed among several different layers of the Hilbert-style theory. Some Euclidean propositions become axioms or immediate consequences of axioms; some receive complete synthetic proofs; some are stated but assigned as exercises; some are represented by more general modern propositions; and some are treated only after numerical measurement has been introduced.

For the present project, this is a significant distinction. Greenberg is an important reference for the mathematical correctness and dependency structure of the Hilbert reconstruction, but it is not a source from which the Lean proofs of I.1–I.20 can simply be transcribed.

.1.4 Consequences for the formal reconstruction

The comparison also clarifies the role of the intermediate Hilbert library developed in the project. A textbook proof can suppress operations that become explicit obligations in Lean: permutations of collinearity, orientations of betweenness, transport along rays, normalization of unoriented angles, transport of segment and angle order under congruence, and repeated proofs of noncollinearity.

Thus the formal reconstruction has two distinct source levels:

1. the classical mathematical argument, for which Greenberg and related literature provide reference results and proof ideas;
2. the explicit synthetic proof infrastructure required to make those arguments executable in Lean.

The second level is represented by the project’s Book Zero and Hilbert interface. It should not be confused with a claim that the underlying Euclidean theorems are new. Its purpose is to expose and organize the elementary proof operations that conventional mathematical prose routinely leaves implicit.

.1.5 Current checkpoint

At the present checkpoint, Euclid Book I, Propositions I.1–I.20 have been reconstructed in the project. The Greenberg comparison should therefore be read as a source audit for this completed initial block, not as a claim about the status of the remaining propositions of Book I. Further sources can be added to the comparison independently as the literature review proceeds.

Appendix A

Hartshorne: Equal Content and the Algebra of Areas

A.1 Purpose and scope

This appendix records a source audit of Euclid Book I, Propositions I.1–I.20, against Robin Hartshorne’s *Geometry: Euclid and Beyond* [5].

The audit has a specific purpose. Hartshorne begins with Euclid’s own proofs and constructions, analyzes where their logical structure is incomplete, and then reconstructs the relevant part of Book I inside a Hilbert-plane framework. For each proposition I.1–I.20, we record:

- whether Hartshorne regards Euclid’s proof as valid as it stands;
- whether the proposition is replaced by an axiom or by a more general Hilbert-plane theorem;
- whether Hartshorne supplies a complete proof, a proof sketch, or an exercise;
- which hidden order, betweenness, separation, congruence, or existence issue he explicitly identifies;
- which points are especially relevant to a fully formal synthetic reconstruction.

The purpose is not to claim novelty for the classical mathematics. Rather, the audit identifies the exact level at which a conventional synthetic proof still leaves obligations that must be made explicit in a formal proof assistant.

A.2 Hartshorne’s organization of the problem

Hartshorne separates the discussion into two stages. Chapter 1 studies Euclid’s own geometry and repeatedly asks whether a proof depends on a diagram, an unstated positional assumption, or a construction whose existence has not yet been justified. Chapter 2 then introduces Hilbert’s axioms of incidence, betweenness, and congruence and reconstructs the early propositions inside that framework.

The key structural statement appears in the section on Hilbert planes: apart from I.1 and I.22, the results of Book I up to that point can be recovered in an arbitrary Hilbert plane. Proposition

I.1 requires an additional intersection or continuity assumption, and I.22 is another exception. This makes I.1–I.20 a particularly natural block for testing the Hilbert reconstruction.

A.3 Comparison table

Table A.1: Euclid I.1–I.20 in Hartshorne’s Hilbert-plane development

Euclid	Hartshorne counterpart	Status in Hartshorne	Source-audit remarks and relevance to formalization
I.1	Construction of an equilateral triangle on a given segment.	Hartshorne treats I.1 as the first genuine exception to the Hilbert-plane recovery of Book I. The intersection of the two circles used by Euclid cannot be derived from the basic Hilbert-plane axioms alone.	Chapter 1 already isolates the hidden existence problem: the diagram shows two circles meeting, but Euclid’s stated postulates do not guarantee that they do. Hartshorne therefore requires an additional intersection or continuity principle. This is a foundational dependency, not a minor gap in presentation.
I.2	Transporting a given segment to a specified point.	Hartshorne states that I.2 is effectively replaced by the segment congruence construction axiom C1.	The Euclidean proposition becomes primitive infrastructure in the Hilbert system. This is important for later segment arithmetic and for the synthetic interpretation of “adding” segments by extending a ray and laying off a congruent copy.
I.3	Cutting off from a greater segment a segment equal to a given smaller segment.	Hartshorne groups I.3 with I.2 as effectively replaced by C1 together with the order theory of segments.	The proposition is no longer structurally fundamental as a named theorem. Its mathematical content is decomposed into segment construction, uniqueness, betweenness, and segment ordering.
I.4	SAS triangle congruence.	SAS is taken as axiom C6. Hartshorne analyzes Euclid’s superposition argument and concludes that it relies on an unstated theory of rigid motions.	This is one of the sharpest differences between Euclid’s proposition order and a Hilbert-style dependency order. The proof by superposition is not accepted as a derivation from Euclid’s explicit postulates and common notions.
I.5	Base angles of an isosceles triangle.	Hartshorne explicitly reinterprets Euclid’s proof in the Hilbert system and gives the proof in detail using segment construction and SAS.	This is a direct example of the program of recovering Euclid inside a Hilbert plane. Hartshorne also points out the hidden betweenness facts needed to justify some of Euclid’s positional claims.
I.6	Converse of the isosceles base-angle theorem.	Hartshorne says that Euclid’s argument is essentially valid, but one point must be rewritten using the formally defined order relation on angles.	The phrase “less than” cannot be left intuitive. The proof depends on having an explicit angle-order relation and on using its asymmetry and trichotomy correctly. This is precisely the kind of obligation that becomes visible in Lean.

Continued on next page

Table A.1 – continued from previous page

Euclid	Hartshorne counterpart	Status in Hartshorne	Source-audit remarks and relevance to formalization
I.7	Uniqueness configuration for triangles on the same base and on the same side.	Hartshorne identifies a hidden dependence on the relative position of the lines and points. The missing betweenness justification is assigned for completion in the Hilbert-plane exercises.	This proposition is a canonical example of a proof whose diagram suppresses case distinctions. Hartshorne stresses that one must know where the lines and intersection points lie before the angle inequalities used in the proof are legitimate.
I.8	SSS triangle congruence.	Hartshorne replaces Euclid's superposition proof with a new theorem, Proposition 10.1 (SSS), proved from the Hilbert congruence axioms.	A complete Hilbert-style proof is supplied. The construction creates a copy of one triangle on the opposite side of a side of the other and uses angle addition and SAS. The positional cases are explicitly noted.
I.9	Construction of an angle bisector.	Hartshorne reconstructs I.9 using Hilbert tools. Since I.1 is not available in an arbitrary Hilbert plane, he replaces Euclid's equilateral-triangle step by the construction of an isosceles triangle.	He explicitly notes that it is not enough to perform the construction: one must prove that the constructed ray really lies in the interior of the original angle. That requires the order/separation machinery.
I.10	Construction of the midpoint of a segment.	Hartshorne reconstructs the proposition by the same strategy: replace Euclid's use of I.1 by an isosceles triangle. The corresponding Hilbert-tools construction is left as Exercise 10.2.	The proposition is therefore recoverable in a Hilbert plane without using the circle-intersection principle required by I.1. This is an important alternative dependency route.
I.11	Erecting a perpendicular at a point of a line.	Hartshorne explains that the construction can be recovered using Hilbert tools and assigns the explicit construction as Exercise 10.3.	The reconstruction also establishes the existence of right angles, which is not taken for granted beforehand. The proof is therefore part of the basic congruence-and-angle infrastructure rather than a compass construction in Euclid's original sense.
I.12	Dropping a perpendicular from a point to a line.	Hartshorne explicitly says that Euclid's original compass construction does not work in an arbitrary Hilbert plane. A new existence proof is required; the explicit construction is Exercise 10.4.	This is a significant foundational note. The failure is not that the theorem is false, but that Euclid's chosen construction imports an intersection principle unavailable in the bare Hilbert system.
I.13	Angles formed when a straight line stands on another straight line.	Hartshorne replaces the Euclidean proposition by Proposition 9.2 on supplements of congruent angles.	He remarks explicitly that Proposition 9.2 may be viewed as the proper Hilbert-style replacement for I.13. The modern statement is more general and fits directly into the angle-congruence infrastructure.

Continued on next page

Table A.1 – continued from previous page

Euclid	Hartshorne counterpart	Status in Hartshorne	Source-audit remarks and relevance to formalization
I.14	Converse straight-line criterion from adjacent supplementary angles.	Hartshorne treats I.14 as an easy consequence of the preceding supplement theory and asks for a careful Hilbert-plane formulation in Exercise 10.7.	The theorem is mathematically routine only after the modern notions of rays, supplementary angles, and angle congruence have been made precise. The formal version therefore belongs naturally to the low-level angle interface.
I.15	Vertical angles are equal.	Corollary 9.3: vertical angles are congruent. A complete short proof is given from the theory of supplementary angles.	This is a clean example in which Hartshorne replaces Euclid’s common notion argument by an explicitly stated congruence theorem.
I.16	Exterior Angle Theorem.	Proposition 10.3 gives a full Hilbert-plane proof.	Hartshorne returns to the exact point he had criticized in Chapter 1: the constructed ray must really lie inside the relevant angle. He closes this gap using betweenness and plane separation. Thus the proof makes explicit the hidden positional content of Euclid’s diagram.
I.17	Any two angles of a triangle are together less than two right angles.	Hartshorne says that I.17–I.21 are valid in a Hilbert plane after reinterpretation. For I.17 he avoids treating “the sum of two angles” as a primitive object and reformulates the statement using supplements and angle order.	This is especially relevant to formalization: the familiar informal statement must be replaced by a proposition expressible in the available angle language.
I.18	The greater side of a triangle subtends the greater opposite angle.	Hartshorne regards the Euclidean argument as recoverable but assigns a careful Hilbert-plane proof as Exercise 10.6.	The exercise explicitly calls attention to betweenness and inequalities. These are exactly the two classes of hidden obligations that dominate a formal reconstruction of I.18.
I.19	The greater angle of a triangle subtends the greater opposite side.	Included in Hartshorne’s statement that I.17–I.21 are valid in a Hilbert plane after the preceding reconstruction of order and congruence.	Hartshorne does not single out a new main-text proof comparable to Proposition 10.3. Its validity rests on the now-formalized order relations for sides and angles and on the preceding comparison theorem.
I.20	Triangle inequality.	Hartshorne says that I.17–I.21 are valid in a Hilbert plane and assigns a careful proof of I.20 as Exercise 10.8.	The theorem therefore belongs to neutral Hilbert geometry. Hartshorne’s exercise status is important: the mathematics is declared recoverable, but the complete chain of synthetic justifications is left to the reader. A formal proof must expose all those omitted steps explicitly.

A.4 Foundational observations retained from Hartshorne

Several remarks in Hartshorne are important beyond the immediate I.1–I.20 audit and should be preserved for later work.

- Hartshorne repeatedly uses a proof from a diagram as a warning sign. When a step depends on the relative position of points, lines, or rays, that position must be derived from the axioms rather than read from the figure.
- Betweenness and plane separation are not secondary bookkeeping devices. They are the mechanism that repairs several of the early Euclidean proofs, especially I.7, I.9, and I.16.
- Hartshorne’s Hilbert plane is built from incidence, betweenness, and congruence. The parallel postulate is kept separate so that one can identify which propositions belong to neutral geometry.
- The first major exception is I.1. The basic Hilbert-plane axioms do not guarantee the intersection of the two circles used by Euclid. Hence an additional intersection/continuity principle is required.
- Euclid’s constructions and Hilbert’s existence axioms should not be conflated. A Euclidean compass construction may fail as a proof in an arbitrary Hilbert plane even when the corresponding geometric object can still be proved to exist by another route.
- Hartshorne replaces Euclid’s superposition method by axiomatic SAS. He interprets superposition as requiring a theory of rigid motions, which is not supplied by Euclid’s stated postulates.
- Segment addition is formalized synthetically. Hartshorne defines the sum of two segments by laying off a congruent copy on a ray and proves that this construction is well-defined up to congruence.
- Segment order is then defined using betweenness and congruence, and Hartshorne proves its compatibility with congruence and its trichotomy properties.
- The angle theory is developed in parallel: supplementary angles, vertical angles, angle addition, and angle order are all made explicit before the later Euclidean propositions are recovered.
- I.17 illustrates an important language issue. An informal sum of angles need not correspond directly to a primitive formal object, so the theorem is reformulated using inequalities and supplements.
- Hartshorne explicitly identifies I.22 as another exception beyond the current checkpoint. Thus the continuity/intersection issue encountered at I.1 returns immediately after I.20 and is likely to matter in the next phase of Book I.

A.5 Result of the Hartshorne audit

Hartshorne does not present Euclid I.1–I.20 as a sequence of twenty independent modern theorems. Instead, he reorganizes the material according to the logical structure of a Hilbert plane.

Some Euclidean propositions become axioms or primitive constructions (I.2–I.4); some receive complete Hilbert-style proofs (notably I.5, I.8, I.15, and I.16); some are reconstructed by replacing

Euclid’s original construction with one valid in a Hilbert plane (I.9–I.12); and several are declared recoverable but are left to the reader as careful exercises (including I.18 and I.20).

The most important feature of the source for formal work is not merely which theorem is available. Hartshorne repeatedly identifies the exact kind of hidden obligation that must be supplied: betweenness, side-of-line information, angle order, segment order, interior-of-angle claims, and existence of intersections.

For a proof assistant, these are not editorial details. They are explicit proof terms and intermediate lemmas. Hartshorne therefore provides an unusually close mathematical guide to the level of granularity required by a fully formal synthetic reconstruction, even where the final derivation is still left as an exercise.

A.6 Current checkpoint

This appendix concerns only Euclid Book I, Propositions I.1–I.20. Observations relevant to later propositions have been retained when Hartshorne explicitly connects them to the present material, but no claim is made here about the complete status of the remainder of Book I.

A.7 Book II semantic checkpoint: Proposition II.5

The preceding proposition-by-proposition audit concerns Book I. For the present Book II reconstruction, Hartshorne has a different role: his treatment of equal content supplies semantic control for the algebraic-looking identities of plane figures. Proposition II.5 does not require numerical area or multiplication of segment lengths in order to state its equality of figures; the familiar algebraic identity is only a modern interpretive shadow of the synthetic configuration [5].

For the current Lean theorem the relevant distinction is between this general semantic reading and the stronger certificate actually constructed by the formal proof. The configured II.5 theorem concludes an exact `HilbertScissorsEq`, obtained from exact rectangle dissections, representative transport, formal addition, and transitivity. That exact finite-scissors strengthening is a fact about the project architecture and should not be attributed automatically to Hartshorne’s general equal-content relation.

Thus the durable Hartshorne-side observation at II.5 is unchanged in kind from the opening of Book II: the identities may be read synthetically as equalities of content between figures, while numerical area remains unnecessary. What is new in the formal proof is the way earlier Book II exact-scissors results are composed to realize that identity.

Appendix B

Hilbert: Content and Segment Arithmetic

B.1 Purpose and scope

This appendix records the Hilbert-side source audit for Euclid Book II at the current checkpoint, Propositions II.1–II.5. Hilbert is relevant in two distinct ways: Chapter IV supplies the synthetic theory of plane content, whereas Section 15 later develops an arithmetic of segments. The documentary purpose is to keep those layers separate and to identify which one is actually used by the current formal reconstruction.

B.2 Hilbert’s organization of the relevant theory

Hilbert’s Chapter IV distinguishes finite equidecomposability from the more general relation of equicomplementability. The theory of plane area begins on the basis of Groups I–IV and does not need to be introduced as a numerical area function [2].

For the present project the useful hierarchy is

exact finite dissection $\implies$ equicomplementability $\implies$ later numerical area semantics,

with no converse inserted merely for convenience.

Section 15 is a different layer. Hilbert there defines segment addition and a geometric product, using a fixed unit segment and parallel construction, and then proves the familiar algebraic laws, including commutativity and distributivity [2]. That later arithmetic is a valuable control source precisely because it shows what would be required if Book II were to be formalized through segment multiplication. The current Lean development does not take that route.

B.3 Comparison table

Table B.1: Euclid II.1–II.5 against Hilbert’s content and segment-arithmetic layers

Euclid	Hilbert counterpart	Status in the Grundlagen	Source-audit remarks and relevance to formalization
II.1	Chapter IV finite decomposition and content theory; later Section 15 distributivity of segment multiplication is a structural analogue.	No one-to-one numbered Hilbert theorem reproducing Euclid II.1 was located in the current audit. The relevant material is organized by content theory and, later, by segment arithmetic rather than by Euclid’s Book II numbering.	The formal proof belongs to the earlier content/dissection layer. Additivity is obtained from an actual rectangle cut and representative transport, not by invoking the later distributive law for a segment product.
II.2	Chapter IV content plus the geometric symmetry of a rectangle; Section 15 commutativity is only a later arithmetic analogue.	Again the source correspondence is structural rather than proposition-level. Hilbert’s later commutative segment multiplication must not be read backward into the scissors proof.	The Lean theorem makes the distinction concrete: the required exchange of the two containing sides is implemented by cyclic rotation of a rectangle representative, preserving exact scissors content.
II.3	Chapter IV content/dissection; later Section 15 distributivity gives the arithmetic pattern only after segment multiplication has been defined.	No separate proposition-level Hilbert counterpart was located. The source supports the distinction between geometric content operations and the later arithmetic of segments.	II.3 is a direct specialization of the binary II.1 cut with the chosen segment used as the fixed side. The square is packaged as a rectangle contained by the same segment twice. No segment product and no side-order transport are needed.
II.4	Chapter IV finite content composition; later Section 15 distributivity and commutativity give an arithmetic analogue of the square expansion.	No one-to-one Hilbert proposition corresponding to Euclid II.4 was located in the current audit. Hilbert’s source role remains architectural.	The formal theorem composes exact II.2 with two exact II.3 instances and a geometric rectangle rotation. Despite the modern shadow $(a + b)^2 = a^2 + b^2 + 2ab$, no segment product, unit segment, or numerical area is used.
II.5	Chapter IV content/dissection; Section 15 segment multiplication supplies only a later arithmetic analogue of the identity.	No one-to-one Hilbert proposition corresponding to Euclid II.5 was located in the current audit. The source role remains the separation of synthetic content from the later arithmetic of segments.	The formal theorem combines midpoint and order information with exact II.1 and II.3 rectangle decompositions, representative transport, and scissors addition. The modern shadow $(a - x)(a + x) + x^2 = a^2$ is not used as a proof object; no segment product, unit segment, numerical area, or cancellation axiom is introduced.

B.4 Structural observations retained from Hilbert

The following distinctions should be preserved as Book II grows.

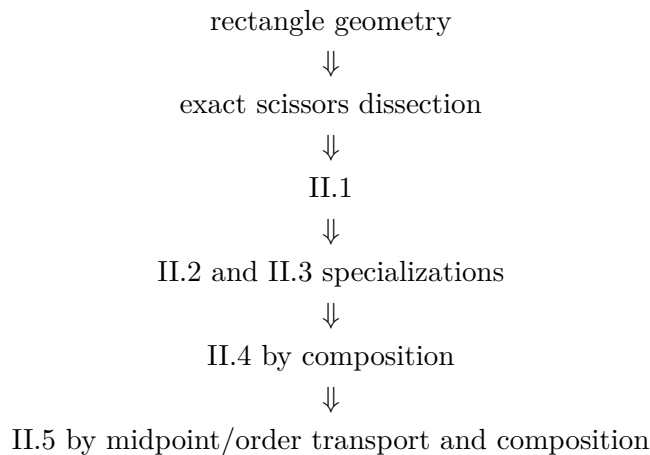
- Plane content and segment arithmetic are separate theories in the Grundlagen and should remain separate in the documentation.

- Exact finite dissection is stronger than the general content relation; when Lean proves `HilbertScissorsEq`, that fact comes from the local proof architecture rather than from a blanket identification of all equal-content figures with equidecomposable ones.
- Segment multiplication is not primitive. It requires a unit segment, proportion geometry, and parallel constructions before algebraic laws such as commutativity and distributivity can even be stated.
- Consequently, a distributive-looking Euclidean proposition can be formalized below the arithmetic layer when its geometry is literally a finite cut-and-paste argument.

B.5 II.1–II.5: exact dissection below segment arithmetic

The opening five propositions give a useful stress test of the separation. II.1 establishes a reusable exact rectangle split. II.2 shows that the apparent exchange of factors can still be implemented geometrically by rotating a rectangle presentation. II.3 then shows that even this extra symmetry operation is unnecessary when the ordered side data already match the binary II.1 interface. II.4 is the first compositional step: it combines II.2 with two II.3 expansions and uses only exact scissors addition and representation transport. II.5 then adds midpoint and order data and composes the same rectangle/scissors mechanisms into the unequal-parts identity, still without entering segment arithmetic.

The resulting formal chain is therefore



not

segment field $\longrightarrow$ multiplication $\longrightarrow$ algebraic simplification.

B.6 Result of the Grundlagen audit

At the II.5 checkpoint Hilbert’s main contribution is architectural. Chapter IV validates the use of a synthetic content layer independently of numerical area, while Section 15 marks a later arithmetic boundary. The current Book II formalization through II.5 remains on the earlier side of that boundary.

The rectangle API therefore has a clear foundational interpretation: congruent segments and right angles determine representative rectangles, and the scissors layer compares their content. The algebra-like behavior observed in II.1–II.3 is derived from that geometry.

B.7 Current checkpoint

This appendix now concerns Euclid Book II, Propositions II.1–II.5. At this stage the durable Hilbert-side picture is

Groups I–IV geometry $\longrightarrow$ plane-content/dissection layer $\longrightarrow$ rectangle calculus for II.1–II.5,
while Section 15 segment arithmetic remains a later comparison layer rather than a dependency of the proofs.

Appendix C

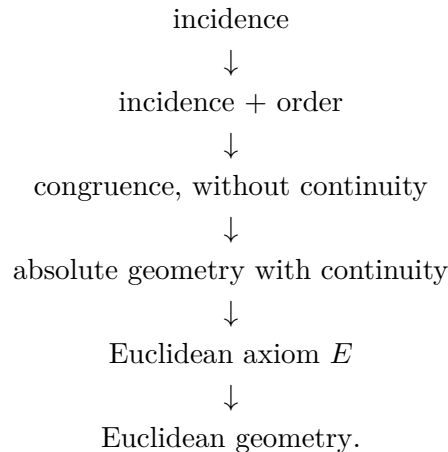
Borsuk–Szmielw: Axiomatic Layers and Representation

C.1 Why this source matters

Karol Borsuk and Wanda Szmielw organize elementary geometry in a way that is unusually close to the architectural questions arising in a formal reconstruction.

Their primitive geometric language consists of points, lines, planes, betweenness, and equidistance. The axioms are divided into five groups: incidence, order, congruence, continuity, and finally either the Euclidean axiom E or the Bolyai–Lobachevskian axiom. The first four groups constitute absolute geometry.

More important for the present purpose, the authors explicitly state that one should study not only which theorems are true, but also which means are necessary to prove them. Their exposition is therefore deliberately layered:



This makes the book especially useful for deciding where a theorem belongs in a formal library. A result corresponding to a Euclidean proposition may turn out to be:

- part of the incidence/order language;
- a consequence of congruence alone;

- available before continuity;
- available only after continuity;
- genuinely Euclidean, requiring E ;
- or absent as a named theorem because Borsuk–Szmielew replace it by a stronger structural result.

The last column of the table below therefore asks a practical question: what does the Borsuk–Szmielew placement suggest for the Hilbert interface and the Book Zero layer?

C.2 Layer notation

For compactness the table uses the following descriptions.

Layer	Meaning
Inc/Ord	Chapter I: incidence and order, including betweenness, half-lines, half-planes, angles as ordered ray configurations, triangles, quadrangles, Pasch-type results, polygons, and triangulation.
Congr-no-Co	Chapter II: congruence added, but the Axiom of Continuity is not used.
Absolute	Chapter III: the Axiom of Continuity is added; the full first four groups of axioms are now available.
Euclidean E	Chapter V: absolute geometry plus the Axiom of Euclid E , equivalent in the book to Euclid’s Fifth Postulate.

C.3 Euclid in the Borsuk–Szmielew development

Table C.1: Euclidean propositions in the axiomatic layers of Borsuk–Szmielew

Euclid	Borsuk–Szmielew counterpart	Layer	Structural significance for the reconstruction
I.1	No one-to-one construction theorem was located. Equilateral triangles are defined in Chapter II, but the audited text does not reproduce Euclid’s two-circle construction as a named theorem.	No direct counterpart	This is a useful absence. Borsuk–Szmielew do not organize the congruence theory by reconstructing Euclid’s first circle construction. The formal project should therefore keep the classical construction DAG separate from the axiomatic layer in which an equilateral-point existence theorem is eventually supplied.
I.2	Axiom C5: on every half-line with origin a , for every segment pq , there exists exactly one point b such that $ab = pq$.	Congr-no-Co	Euclid’s segment transport has become primitive and is strengthened to existence plus uniqueness on a prescribed half-line. This is precisely interface-level segment layoff, not a proposition that later proofs should reconstruct from circles.

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Table C.1 – continued from previous page

Euclid	Borsuk–Szmielew counterpart	Layer	Structural significance for the reconstruction
I.3	Axiom C5 together with segment order and betweenness.	Congr-no-Co	Cutting a prescribed length from a longer segment is absorbed into ray layoff plus order. The natural formal location is a small Book Zero lemma over segment layoff and betweenness.
I.4	Theorem 22(I): two sides and the included angle determine a congruent triangle. The text immediately records this as the SAS congruence criterion.	Congr-no-Co	SAS is developed before continuity. This is strong evidence for keeping it in the neutral congruence interface. Euclid’s superposition argument is not part of the Borsuk–Szmielew proof architecture.
I.5	Theorem 23: in a triangle, congruent sides are opposite congruent angles; in particular, the base angles of an isosceles triangle are congruent.	Congr-no-Co	The isosceles theorem is packaged together with its converse in one biconditional result. The reusable API should reflect this stronger relationship instead of treating I.5 and I.6 as unrelated endpoints.
I.6	The converse direction of Theorem 23: congruent opposite angles imply congruent opposite sides.	Congr-no-Co	Borsuk–Szmielew eliminate Euclid’s separate reductio construction. Once segment and angle congruence are organized properly, I.6 is simply the reverse implication of the same structural theorem as I.5.
I.7	No separate theorem corresponding exactly to Euclid I.7 was located. Its uniqueness content is absorbed into the order theory, unique angle construction (Theorem 13), and the strong congruence axioms.	Congr-no-Co	This is an example of a Euclidean proposition disappearing under a stronger interface. The formal question is not how to imitate I.7, but which general uniqueness and side-of-line lemmas make its role unnecessary.
I.8	Theorem 21: if the three corresponding sides of two triangles are congruent, then the three corresponding angles are congruent.	Congr-no-Co	This is the SSS content in the Borsuk–Szmielew organization, and it is available without continuity and without the Euclidean axiom. For the formal project this strongly supports treating SSS as derived neutral congruence infrastructure rather than Euclidean input.
I.9	Theorems 33–34: a bisector lies between the sides of the angle, and every angle has exactly one bisector.	Congr-no-Co	The source gives a stronger result than Euclid’s bare construction: existence and uniqueness are packaged together. Angle-bisector existence therefore belongs naturally in the reusable interface.
I.10	Theorems 31–32: an equidistant collinear point is between the endpoints, and every segment has exactly one midpoint.	Congr-no-Co	This is especially valuable formally. Borsuk–Szmielew do not take the diagrammatic position of the midpoint for granted; they first prove the betweenness theorem and only then existence and uniqueness. This is exactly the separation needed in Lean.
I.11	Theorem 76: through a point lying on a line there exists exactly one line in the plane perpendicular to the given line.	Congr-no-Co	Perpendicular erection is already a neutral interface theorem. Later propositions should consume it directly rather than replay an auxiliary Euclidean triangle construction.

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Table C.1 – continued from previous page

Euclid	Borsuk–Szmielew counterpart	Layer	Structural significance for the reconstruction
I.12	Theorem 77: through a point outside a given line there exists exactly one perpendicular to that line.	Congr-no-Co	Dropping a perpendicular is likewise available before continuity. The theorem is stronger and cleaner than Euclid’s construction-oriented formulation and belongs naturally in the Hilbert interface.
I.13	Sections 21–24 of Chapter II: addition of free angles, adjacent angles, right angles, and the fact that an angle adjacent to a right angle is right (Theorems 63, 69–71).	Congr-no-Co	Borsuk–Szmielew replace informal angle arithmetic by an explicit calculus of free angles. This is a strong source precedent for using synthetic supplement/addition relations rather than numerical degrees in the formalization.
I.14	Theorem 13 gives unique construction of a congruent angle on a specified side; Theorems 14–15 control adjacent angles and their uniqueness.	Congr-no-Co	The converse straight-line criterion is absorbed into a more general uniqueness theory for angle placement. The formal analogue should be a small collinearity/adjacency consequence of that interface.
I.15	Theorem 16: vertical angles are congruent.	Congr-no-Co	A direct reusable theorem. Its low position in the congruence chapter confirms that vertical-angle congruence should be generic infrastructure for later triangle and parallel arguments.
I.16	Theorem 35: an exterior angle of a triangle is greater than each nonadjacent interior angle.	Congr-no-Co	This is one of the central neutral theorems. The proof explicitly uses midpoint, vertical-angle congruence, SAS, and angle order. It is also the engine for constructing nonintersecting lines before the Euclidean axiom enters.
I.17	A short consequence of Theorem 35 together with the developed theory of adjacent and right angles. It is not necessary to use the later Saccheri–Legendre theorem for this Euclidean proposition.	Congr-no-Co	This distinction prevents an important misclassification. Borsuk–Szmielew later prove stronger angle-sum results after continuity, but Euclid I.17 itself already belongs to the earlier neutral congruence layer.
I.18	Theorem 40, one direction: in a triangle the greater side lies opposite the greater angle.	Congr-no-Co	I.18 and I.19 are unified. Borsuk–Szmielew package strict side and angle comparison into a biconditional theorem, exactly the sort of strong reusable comparison lemma useful in Book Zero.
I.19	Theorem 40, converse direction: the greater angle lies opposite the greater side.	Congr-no-Co	The converse is not a separate construction. It is the reverse direction of the same strict comparison equivalence used for I.18.
I.20	Theorem 50: for distinct points a, b, c , $[ab] + [bc] > [ac]$, with equality exactly when $B(a, b, c)$.	Congr-no-Co	This is substantially stronger than the triangle-only formulation. It combines triangle inequality with the collinear equality case and uses the explicit algebra of free segments. It is an excellent model for a robust segment-order interface.

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Table C.1 – continued from previous page

Euclid	Borsuk–Szmielw counterpart	Layer	Structural significance for the reconstruction
I.21	Theorem 52 generalizes the side-sum part to a convex polygonal line inside a triangle; the angle part is controlled by the Exterior Angle Theorem 35 and the order theory of rays.	Congr-no-Co	The source replaces the one-off proposition by a more general polygonal inequality. For formal work the useful target is therefore the reusable generalized lemma rather than the exact Euclidean diagram.
I.22	No one-to-one prescribed-three-sides construction theorem was located in the audited elementary sequence. The book has the triangle inequality and a strong congruence construction axiom, but does not replay Euclid’s circle-intersection construction here.	No direct counterpart	This should be kept as an explicit gap in the source map. Existence of a triangle with three prescribed side lengths is logically different from SSS uniqueness/congruence and should not be silently identified with Theorem 21.
I.23	Theorem 13: for a given ray, chosen half-plane, and given angle, there exists exactly one ray on that side making a congruent angle.	Congr-no-Co	Euclid’s angle-copy construction becomes a primitive-style transport theorem with uniqueness. This is one of the strongest direct precedents for the angle-construction API already used in the Hilbert route.
I.24	Theorem 41: with two corresponding sides fixed, comparison of the included angles is equivalent to comparison of the third sides.	Congr-no-Co	Borsuk–Szmielw combine the Euclidean comparison theorem and its converse into one biconditional. This is a natural interface theorem for replacing configuration-heavy local proofs.
I.25	The converse direction of Theorem 41.	Congr-no-Co	As with I.18/I.19, the source suggests exposing one symmetric comparison theorem rather than maintaining two large proposition-sized proofs.
I.26	Theorem 22(II): one side and the two angles adjacent to it determine a congruent triangle (ASA).	Congr-no-Co	The ASA part of I.26 is explicit and neutral. The audited text does not package Euclid’s second AAS configuration under a separate theorem number here, so the appendix does not silently identify the two formulations.
I.27	Theorems 37–39. Equal appropriate transversal angles give nonintersection; Theorem 39 concludes that through an external point there is at least one line not intersecting the given line.	Congr-no-Co	This is one of the most important source placements in the book. The angle-to-parallel direction and existence of a nonintersecting line are established before continuity and before the Euclidean axiom.
I.28	A consequence of the same Theorems 37–38 together with the earlier adjacent/vertical-angle normalization theorems.	Congr-no-Co	The two Euclidean angle criteria are wrappers over the neutral nonintersection theorem. The conclusion “parallel” does not by itself make the result Euclidean.

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Table C.1 – continued from previous page

Euclid	Borsuk–Szmielew counterpart	Layer	Structural significance for the reconstruction
I.29	After E , Chapter V develops the Euclidean characterization of parallel lines and Theorem 10: angles whose corresponding sides are parallel are congruent.	Euclidean E	This is the reversal of I.27. Before E , appropriate angle data produce nonintersection; after E , parallelism can be converted back into angle congruence. The source therefore marks the same neutral-to-Euclidean boundary that the formal reconstruction uses.
I.30	Chapter V, Theorem 7: parallelism of lines is transitive.	Euclidean E	Parallel transitivity is explicitly downstream of the Euclidean axiom. It should therefore be classified as Euclidean interface infrastructure, not as a neutral convenience lemma.
I.31	Chapter II, Theorem 39 gives existence of a non-intersecting line. Axiom E plus that theorem gives Chapter V, Theorem 1, uniqueness; Theorem 6 states the corresponding unique parallel through a point.	Existence: Congr-no-Co; uniqueness: Euclidean E	Borsuk–Szmielew make the existence/uniqueness split exceptionally clean. This is a direct architectural model for keeping the two layers separate in Lean.
I.32	Chapter V, Theorem 2: the sum of the angles of every triangle is π .	Euclidean E	The equality form of the triangle angle-sum theorem is placed immediately after the Euclidean axiom. A synthetic Lean statement need not use numerical measure, but its dependency must remain Euclidean.
I.33	Chapter V, Theorem 12, sufficiency direction: a plane quadrangle is a parallelogram when one pair of corresponding directed sides is parallel and the corresponding segments are congruent.	Euclidean E	This theorem is stronger and more useful than a literal copy of I.33. It characterizes a parallelogram from exactly the sort of parallel-plus-congruence data that arise in I.33. It is a strong candidate for reusable interface infrastructure.
I.34	Chapter V, Theorem 12, necessity direction and proof: in a parallelogram the diagonal triangles are congruent, yielding equality of opposite sides and the corresponding angle data.	Euclidean E	The proof architecture is almost tailor-made for the formal I.34 interface: parallel half-lines give angle congruences, then triangle congruence supplies opposite-side data. I.34 should expose this package for immediate downstream use.
I.35	No Euclidean equal-content or equicomplementability theorem corresponding directly to I.35 was located. Chapter I already has a substantial theory of polygons and triangulation, while Chapter V has the parallelogram theory needed immediately before I.35.	No direct counterpart	The absence is informative. Borsuk–Szmielew supply rigorous figure/triangulation infrastructure and Euclidean parallelogram infrastructure, but they do not insert a Hilbert/Hartshorne-style equal-content layer at this point. Thus they are highly useful for the components feeding I.35, but not the primary source for the semantics of its conclusion.

C.4 The strongest architectural correspondences

The most useful feature of the Borsuk–Szmielw source is that several Euclidean propositions are replaced by stronger interface theorems.

C.4.1 Transport is primitive-strength infrastructure

Axiom C5 is stronger than the bare construction in I.2: it gives exactly one point on a chosen half-line at a prescribed distance. Theorem 13 is the angular analogue: on a chosen side of a ray, an angle congruent to a given angle can be constructed uniquely.

Thus two operations that appear relatively late or require elaborate construction in Euclid become basic transport mechanisms:

segment layoff and angle layoff.

This is directly consonant with the Hilbert-style formal interface.

C.4.2 Pairs of Euclidean propositions become biconditionals

Theorem 23 unifies I.5 and I.6:

equal sides $\iff$ equal opposite angles.

Theorem 40 unifies I.18 and I.19:

greater side $\iff$ greater opposite angle.

Theorem 41 similarly unifies I.24 and I.25 when two corresponding sides are fixed:

greater included angle $\iff$ greater third side.

For library design, these are more informative than the proposition numbers. They identify the natural reusable theorem boundaries.

C.4.3 Midpoints and bisectors include positional information

The midpoint discussion is especially close to the needs of formal geometry. Borsuk–Szmielw first prove that a collinear point equidistant from the endpoints of a segment must lie between them. Only then do they prove existence and uniqueness of the midpoint.

The angle-bisector discussion has the same pattern: first the bisector is shown to lie between the sides of the angle, then existence and uniqueness are established.

The lesson for formalization is precise. “Midpoint” and “bisector” should not be treated merely as metric equalities; their order/ray placement is part of the reusable theorem package.

C.5 The neutral parallel theory is already large

The source is particularly valuable around Euclid I.27–I.31.

From the Exterior Angle Theorem and the congruence/order theory, Borsuk–Szmielew prove that an appropriate equality of transversal angles forces two lines to be disjoint. They then obtain Theorem 39:

through an external point there exists at least one nonintersecting line.

This theorem belongs to Chapter II, before continuity and before the Euclidean axiom.

Only much later, in Chapter V, is Axiom *E* added:

through an external point there is at most one nonintersecting line.

Together the two statements give uniqueness. Chapter V then derives the Euclidean parallel theory: characterization of disjoint coplanar lines, uniqueness of the parallel, transitivity, angles with parallel sides, and parallelograms.

This produces a clean formal boundary:

angle criterion $\Rightarrow$ nonintersection	before <i>E</i> ,
existence of a nonintersecting line	

while

uniqueness of the parallel	after <i>E</i> .
parallelism $\Rightarrow$ angle congruence	
parallel transitivity	
triangle angle sum	

For the present reconstruction this is one of the strongest independent checks on the placement of I.27, I.29, I.30, I.31, and I.32.

C.6 Theorem 12 of Chapter V as an interface theorem

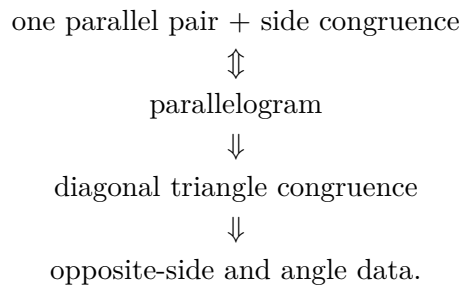
The parallelogram theorem deserves special attention.

Borsuk–Szmielew define a parallelogram in the expected way, by two pairs of opposite sides lying on parallel lines. Their Theorem 12 then proves a necessary and sufficient condition using only one pair of directed parallel sides and equality of the corresponding segments.

In the necessity direction, the proof starts from a parallelogram, transports angles using parallel half-lines, and obtains congruence of the two triangles cut by a diagonal. Opposite-side congruence follows.

In the sufficiency direction, one parallel pair together with one segment congruence forces the remaining pair to be parallel.

This theorem simultaneously explains why I.33 and I.34 should sit close to the reusable Euclidean parallelogram interface:



It is therefore not merely a historical counterpart. It is a plausible API boundary for subsequent formal work.

C.7 What the book contributes at I.35

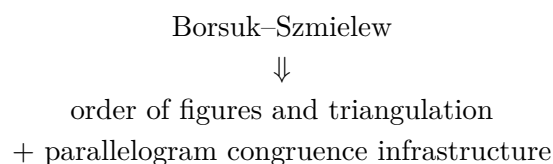
The transition at I.35 is revealing because the source provides two sides of the problem but not the final semantic relation.

Chapter I develops polygons, intersections of polygons with half-planes, and triangulation. This is a rigorous combinatorial theory of plane figures built very low in the axiomatic hierarchy.

Chapter V develops the Euclidean parallelogram theory and supplies the congruence infrastructure immediately preceding I.35.

What is not developed there is a Euclidean theory of equal content or equicomplementability comparable to Hilbert Chapter IV or Hartshorne Section 22. Later references to area in the Bolyai–Lobachevskian part do not supply such a replacement; the authors explicitly say that they will not develop a general theory of area there.

For the formal reconstruction the source therefore has a sharply defined role:



but

Hilbert/Hartshorne remain the control sources for equicomplementability/equal content.

This is not a weakness of the audit. It tells us exactly where one source stops and another must take over.

C.8 Consequences for the formal library

The Borsuk–Szmielw audit suggests a practical classification of reusable results.

C.8.1 Hilbert interface candidates

The following belong naturally to a general synthetic interface rather than to individual Euclidean proposition files:

- unique segment layoff on a ray;
- unique angle layoff on a chosen side;
- SAS and SSS-style triangle congruence;
- isosceles theorem and converse;
- midpoint existence, uniqueness, and betweenness;
- angle-bisector existence, uniqueness, and ray order;
- vertical angles;
- Exterior Angle Theorem;
- strict side/angle comparison;
- perpendicular existence and uniqueness;
- neutral angle criteria for nonintersection.

C.8.2 Book Zero candidates

The developed calculus of free segments and free angles also points to a natural macro-language:

- segment less-than and trichotomy;
- transport of order under congruence;
- segment addition and subtraction;
- triangle inequality with the collinear equality case;
- angle less-than and trichotomy;
- angle addition and adjacency;
- supplement/right-angle normalization.

These are exactly the sorts of operations that prevent later proposition files from expanding into repeated low-level incidence and order proofs.

C.8.3 Euclidean interface candidates

After Axiom *E*, the natural reusable layer contains:

- unique parallel through a point;
- characterization of Euclidean parallels;
- parallel transitivity;
- congruence of angles with respectively parallel sides;
- Euclidean triangle angle sum;
- the Chapter V parallelogram characterization.

The source thus gives not just another list of theorem counterparts, but a plausible decomposition of the formal library itself.

C.9 Conclusion

Borsuk and Szmielew are particularly valuable for the reconstruction because their book asks almost the same structural question as the formal project: not only whether a theorem is true, but which axiomatic resources are actually needed to prove it.

Their treatment makes several boundaries unusually visible. Much of the elementary order and congruence theory, including the Exterior Angle Theorem, midpoints, bisectors, perpendiculars, triangle comparison, and existence of nonintersecting lines, is developed before continuity. The Axiom of Euclid then contributes the uniqueness needed to obtain the full Euclidean theory of parallels. Their parallelogram theorem packages the material around I.33–I.34 in a form that is especially attractive as formal interface infrastructure.

At I.35 the source becomes useful in a different way. Its early theory of polygons and triangulation and its later parallelogram theory provide important components of the proof, while the semantic theory of equal content is left to other sources.

The resulting appendix should therefore be read as an axiomatic-layer map and an API-design aid, not merely as another historical comparison with Euclid.

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